

LOCAL EXCLUSION OF RESIDUAL TYPE I NAVIER–STOKES SEREGIN LIMITS

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ABSTRACT. We assemble the refined residual analysis for Type I Navier–Stokes Seregin limits. A finite-energy Type I singularity produces, through the Seregin blow-up procedure, a nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation carrying positive local velocity L^3 -mass. After the previously treated small, stationary L^3 , uniformly L^3 -tight, and closed structured/decay alternatives have been removed, the remaining residual class is divided into four analytic classes: axisymmetric bounded-circulation profiles, rotational relative equilibria, degenerate stationary-hull profiles with retained local concentration, and the generic terminal residual.

The first class is excluded by a centered angular-circulation equation and an axis-absorption Liouville theorem. The second class is closed by a local Coriolis-flux identity and a local concentration decomposition: a superthreshold rotational flux gives a nontrivial recentered local profile, while a subthreshold flux remains inside the compact local limiting equation. The third class is reduced, by scale reselection for time-translation hulls, to a stationary profile in a degenerate stationary family which occurs as a Seregin limit and still carries local CKN concentration; it is then excluded by the terminal local concentration theorem. The final class is reduced by covariant tail normalization, active-locus extraction, diffuse-defect compactness, critical-tail compactification, active successor path-space recurrence, exclusion of finite separated concentration profiles, indecomposable-profile sequence- L^3 completeness, parasitic-free mildness inheritance, and the Albritton–Barker endpoint Liouville theorem. The paper distinguishes statements imported from Papers I–III, local compactness and bookkeeping statements proved here, and the remaining realized critical-tail rigidity theorem needed to close the critical boundary compactification. No global Coriolis–Leray Liouville theorem for bounded profiles or global tail estimate is used in the residual closure.

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1. INTRODUCTION AND MAIN THEOREM

We write the incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$ as

$$(1.1) \quad \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0.$$

The residual analysis in this paper begins after the standard Type I blow-up construction has been performed. Thus the basic solution is a smooth bounded ancient solution of the centered equation

$$(1.2) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0,$$

on $\mathbb{R}_y^3 \times \mathbb{R}_\tau$. The equation is obtained from the physical ancient representative (U, Q) by

$$(1.3) \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t}U(x, t), \quad P(y, \tau) = (-t)Q(x, t),$$

where $t < 0$ and pressures are always understood modulo addition of a function of time. Conversely,

$$(1.4) \quad U(x, t) = (-t)^{-1/2}V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0.$$

The purpose of the paper is to present the refined residual decomposition as a single standard PDE argument, using only the reductions already established in the preceding papers of the series. To keep the logical status clear, we first state the imported Paper 0 and Paper I–III dependencies. Apart from those four companion papers, the only external inputs used below are the cited peer-reviewed Navier–Stokes regularity and Liouville theorems. All rotational-flux estimates, terminal local concentration decompositions, exclusions of local compactness failure and diffuse exterior concentration, and finite separated-profile exclusions used below are proved in the present paper from local estimates rather than imported as global estimates.

Definition 1.1 (Type I convention used in the chain). In this paper and in the final Papers 0–IV assembly, a terminal singular point $z_* = (x_*, T)$ is called *local pointwise Type I* if there are $\rho > 0$ and $M < \infty$ such that

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

This is the local pointwise Type I hypothesis used in Paper 0 and in [Section 1.1](#). The paper does not use the phrase “Type I” to mean an unrelated global L^∞ , local CKN, weak- L^3 , or Serrin scale condition unless that condition is explicitly stated.

Remark 1.2 (Translation table for standard Type I hypotheses). The global pointwise condition

$$\sup_{t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty$$

implies [Theorem 1.1](#) at every terminal point. Any stronger local criterion that yields the displayed L^∞ envelope on a neighborhood of x_* also enters the same pipeline. Purely CKN-type, weak- L^3 , or other critical criteria are not silently identified with this hypothesis; they require their own implication into the local pointwise envelope before the present exclusion applies.

- (1) Local concentration entry: depends on CKN theory and Paper 0; status: proved/imported.
- (2) Seregin extraction and parasitic-free normalization: depends on Paper I and the Paper II bounded mild formulation; status: proved/imported.
- (3) Small and stationary L^3 strata: depend on Paper I; status: proved/imported.
- (4) Uniformly tight stratum: depends on Paper II and endpoint Liouville input; status: proved/imported.
- (5) Closed structured/decay strata: depend only on the unconditional part of Paper III; status: proved/imported.
- (6) Axisymmetric and rotational residual strata: proved in this paper.
- (7) Terminal active-locus stratification, active path-space recurrence, and diffuse-defect compactness: proved in this paper.
- (8) Realized critical-tail rigidity for the boundary compactification: explicit local state-space input [Theorem 8.21](#).
- (9) Residual indecomposable-profile closure: proved in this paper, with the final sequence- L^3 endpoint supplied by Albritton–Barker.

The final local pointwise Type I exclusion does not use the conditional controlled-swirl, weighted-velocity, Burgers-vortex association, or shell-control hypotheses from Paper III. Those conditional statements remain background criteria for other structured classes and are not on the main logical path closed here.

Theorem 1.3 (Paper I Type I Seregin extraction theorem). *A finite-energy Type I singular point produces a normalized Seregin ancient limit (V, P) satisfying (1.2), with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M < \infty,$$

local suitability and local pressure-gauge compactness, and a compact parabolic cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ for which

$$(1.5) \quad \iint_K |V|^3 dy d\tau \geq \eta_0 > 0.$$

The collection of all normalized Seregin limits generated by the original solution is denoted by $\mathcal{S}$.

Proof. This is Paper I’s extraction theorem [2], including velocity nonvanishing, local suitability, and local pressure compactness. The underlying positive local concentration input comes from Paper 0 [1]. The present paper uses this as an imported reduction and does not reprove the Type I blow-up compactness theorem. $\square$

Definition 1.4 (Imported earlier alternatives). Within a fixed normalized Seregin collection $\mathcal{S}$, the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes are the non-residual alternatives defined in Paper I. The small-amplitude and stationary L^3 alternatives are closed in Paper I; the uniformly tight class is excluded in Paper II [3]; and the closed structured/decay class is precisely the part of the structured alternatives proved empty in Paper III [4]. Any structured profile not belonging to that closed Paper I class is residual by definition and is handled by the present paper. The present paper does not reprove those earlier alternatives and does not add any hypothesis to their statements.

Definition 1.5 (Coarse residual class). Let I, J be the index sets used in the base Type I residual criterion. The coarse residual class $\mathcal{R}(\mathcal{S}; I, J)$ is the complement in $\mathcal{S}$ of the already removed alternatives in [Theorem 1.4](#).

Definition 1.6 (Refined residual decomposition). The refined residual decomposition is the ordered decomposition of the remaining coarse residual class into the following alternatives, each understood after all earlier alternatives and all previously closed non-residual classes have been removed:

$$\mathcal{C}_{\text{ax}}, \quad \mathcal{C}_{\text{rot}}, \quad \mathcal{C}_{\text{stat}}^{\text{ret}}, \quad \mathcal{C}_{\text{gen}}.$$

Here $\mathcal{C}_{\text{ax}}$ is the axisymmetric bounded-circulation class, $\mathcal{C}_{\text{rot}}$ is the rotational relative-equilibrium class, $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is the degenerate stationary-hull class with retained local concentration, and $\mathcal{C}_{\text{gen}}$ is the generic terminal residual class.

Theorem 1.7 (Refined residual closure by local concentration analysis). *Assume the realized critical-tail rigidity input [Theorem 8.21](#). Relative to the imported Paper I–III alternatives in [Theorems 1.3](#) and [1.4](#), one has*

$$\mathcal{C}_{\text{ax}} = \emptyset, \quad \mathcal{C}_{\text{rot}} = \emptyset, \quad \mathcal{C}_{\text{stat}}^{\text{ret}} = \emptyset, \quad \mathcal{C}_{\text{gen}} = \emptyset.$$

Consequently the coarse residual class $\mathcal{R}(\mathcal{S}; I, J)$ is empty.

Proof. The axisymmetric bounded-circulation class is excluded by [Theorem 3.3](#). The rotational relative-equilibrium class is excluded by [Theorem 4.8](#), which uses the localized Coriolis-flux identity and a terminal local concentration decomposition rather than a whole-space Coriolis–Leray Liouville theorem. The degenerate stationary-hull class with retained local concentration $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is excluded by [Theorem 6.6](#), using the terminal local concentration theorem proved in [Sections 8](#) and [9](#). Finally, the generic terminal residual class $\mathcal{C}_{\text{gen}}$ is excluded by [Theorem 10.2](#). These four ordered exclusions exhaust the refined residual decomposition. $\square$

Corollary 1.8 (Paper I residual input). *Assume the realized critical-tail rigidity input [Theorem 8.21](#). For every normalized Seregin collection $\mathcal{S}$ generated by a Paper 0-admissible finite-energy Type I singularity, the Paper I remaining class is empty.*

Proof. Paper I defines its remaining class as the complement, inside $\mathcal{S}$, of the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes. The notation $\mathcal{R}(\mathcal{S}; I, J)$ used here records the same complement together with the ordered removals used in the residual decomposition; the indices I, J add no analytic assumption. By [Theorem 1.7](#), this class is empty. $\square$

Theorem 1.9 (Local pointwise Type I exclusion after residual closure). *Assume [Theorem 8.21](#). No Paper 0-admissible finite-energy local pointwise Type I singularity exists whose normalized Seregin collection is $\mathcal{S}$.*

Proof. Suppose that a Type I singular point exists. By [Theorem 1.3](#), the Seregin blow-up procedure produces a normalized ancient limit (V, P) satisfying [\(1.2\)](#) and the local velocity nonvanishing condition [\(1.5\)](#). By the Paper I class exhaustion, V is either in one of the non-residual classes recorded in [Theorem 1.4](#) or in $\mathcal{R}(\mathcal{S}; I, J)$. The first alternative is impossible by the corresponding closed inputs: the small and stationary L^3 classes are excluded in Paper I, the uniformly L^3 -tight class is excluded in Paper II, and the closed structured/decay class is excluded in Paper III. The second alternative is impossible by [Theorem 1.7](#). This contradiction proves the theorem. $\square$

1.1. Local Type I singularities enter the terminal concentration package. The purpose of this subsection is to make explicit the local reduction that is implicit in the proof architecture of Papers 0–IV. A finite-energy singular point satisfying the local Type I bound of Paper 0 already supplies the local compactness needed to enter the terminal concentration package. Thus the whole-space no-escape hypothesis used to formulate the Paper I admissible class is not an additional assumption for a locally Type I terminal singularity: the local Type I envelope gives compactness on every fixed terminal cylinder which remains inside the local Type I window after rescaling around the singular point, while concentration attempting to leave every such bounded local window is one of the exterior terminal alternatives. The Paper 0 concentration theorem supplies the retained local L^3 -mass.

For this subsection, if $z_* = (x_*, T)$ and $r_n \downarrow 0$, write

$$u_n(x, t) := r_n u(x_* + r_n x, T + r_n^2 t), \quad p_n(x, t) := r_n^2 p(x_* + r_n x, T + r_n^2 t).$$

Also write $Q_R := B_R(0) \times (-R^2, 0)$ and

$$Q_1(0, 0) = B_1(0) \times (-1, 0).$$

Proposition 1.10 (Local Type I rescalings have Seregin compactness). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that there exist $\rho > 0$ and $M < \infty$ such that*

$$(1.6) \quad \operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

Let $r_n \downarrow 0$, and let (u_n, p_n) be the rescaled sequence above. Then the following hold.

(i) For every compact cylinder

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \quad R, S < \infty, \quad \sigma > 0,$$

there exists $n_0(K)$ such that

$$(1.7) \quad \|u_n\|_{L^\infty(K)} \leq M\sigma^{-1/2}, \quad n \geq n_0(K).$$

(ii) On every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$ there are time-dependent pressure gauges $a_{n,K}(t)$ and a constant $C_K < \infty$ such that, for $n \geq n_0(K)$,

$$(1.8) \quad \|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \leq C_K.$$

The constant C_K may depend on the fixed Leray energy level of the original solution, but no concentration or no-escape assumption enters it.

(iii) After subtracting these local pressure gauges, the family (u_n, p_n) is locally precompact in the standard Seregin topology on $\mathbb{R}^3 \times (-\infty, 0)$. Along every subsequence there is a further subsequence and an ancient suitable weak solution (U, Q) such that, on every compact $K \Subset \mathbb{R}^3 \times (-\infty, 0)$,

$$u_n \rightarrow U \quad \text{strongly in } L^q(K), \quad 1 \leq q < \infty,$$

and

$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}(K).$$

The limit is bounded and satisfies

$$|U(x, t)| \leq M(-t)^{-1/2}, \quad t < 0.$$

(iv) More generally, let $\mathbf{c} = (z_n)$, $z_n = (y_n, \tau_n)$, be a parabolic recentering sequence for (u_n, p_n) with $\tau_n \leq 0$. Assume that $\mathbf{c}$ remains in the local Type I window in the following precise sense: for every $R, S < \infty$, all sufficiently large n satisfy

$$r_n(|y_n| + R) < \rho, \quad T + r_n^2(\tau_n - S) > T - \rho^2.$$

If $\mathbf{c}$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness in the sense of [Theorem 8.3](#), and its recentered fields converge locally to a bounded suitable terminal profile. Recenterings which do not satisfy this local-window condition are, by definition, routed to the exterior or diffuse terminal alternatives rather than to the compact local profile class.

Proof. Fix

$$K = B_R \times [-S, -\sigma] \in \mathbb{R}^3 \times (-\infty, 0).$$

For n sufficiently large,

$$r_n R < \rho/2, \quad r_n^2 S < \rho^2/2.$$

Hence, for a.e. $(x, t) \in K$,

$$x_* + r_n x \in B_\rho(x_*), \quad T + r_n^2 t \in (T - \rho^2, T).$$

The local Type I bound [\(1.6\)](#) gives

$$|u_n(x, t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \leq r_n M (T - (T + r_n^2 t))^{-1/2} = M(-t)^{-1/2} \leq M\sigma^{-1/2}.$$

This proves [\(1.7\)](#). In particular,

$$\operatorname{ess\,sup}_{-S < t < -\sigma} \int_{B_R} |u_n(x, t)|^2 dx \leq |B_R| M^2 \sigma^{-1},$$

and

$$\iint_K |u_n|^3 dx dt \leq |K| M^3 \sigma^{-3/2}.$$

We next record the pressure estimate because it is the only place where a local Type I proof can hide a nonlocal issue. Use the whole-space Leray pressure representative. Fix a cutoff $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R \equiv 1$ on B_{4R} , and write in B_{4R}

$$p_n = q_n + h_n, \quad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$

The local term satisfies

$$\|q_n\|_{L^{3/2}(B_{4R} \times [-S, -\sigma])} \leq C_{R,S,\sigma,M}.$$

For the harmonic part choose the gauge $a_{n,K}(t) = h_n(0, t)$. The standard kernel difference estimate for the Riesz pressure kernel gives, for $x \in B_{2R}$,

$$|h_n(x, t) - a_{n,K}(t)| \leq C_R \sum_{m \geq 2} 2^{-4m} \int_{2^m R < |z| < 2^{m+1} R} |u_n(z, t)|^2 dz.$$

For annuli whose physical image remains in $B_\rho(x_*)$, the local Type I bound gives

$$\int_{2^m R < |z| < 2^{m+1} R} |u_n(z, t)|^2 dz \leq C M^2 \sigma^{-1} (2^m R)^3.$$

The corresponding part of the series is bounded by a geometric sum. For the remaining outer annuli one uses only the finite Leray energy:

$$\int_{\mathbb{R}^3} |u_n(z, t)|^2 dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 dy.$$

If m_n is the first index for which $2^{m_n} R r_n \gtrsim \rho$, then

$$\sum_{m \geq m_n} 2^{-4m} \int_{\mathbb{R}^3} |u_n(z, t)|^2 dz \leq C_{\rho,R} r_n^3 \|u\|_{L^\infty(0,T;L^2)}^2 \leq C_{\rho,R,u}.$$

Thus $h_n - a_{n,K}$ is uniformly bounded on $B_{2R} \times [-S, -\sigma]$, and

$$\|p_n - a_{n,K}(t)\|_{L^{3/2}(B_{2R} \times [-S, -\sigma])} \leq C_K.$$

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder $K' \Subset \mathbb{R}^3 \times (-\infty, 0)$. The already obtained local L^2 , L^3 , and pressure-gauge bounds control every term on the right-hand side, and therefore

$$\iint_K |\nabla u_n|^2 dx dt \leq C_K.$$

This proves (1.8).

The Navier–Stokes equation now bounds $\partial_t u_n$ locally in a negative Sobolev space, for instance in $L_t^{3/2} H_x^{-2}(K)$. Aubin–Lions compactness and the uniform L^∞ bound give strong local convergence of the velocity in every finite L^q , after passing to a subsequence. The pressure bound gives weak $L^{3/2}$ convergence after gauges. Passing to the limit in the distributional equations and in the local energy inequality gives an ancient suitable weak solution (U, Q) . The pointwise Type I bound passes to the limit, so U is bounded on every compact negative-time cylinder. This proves (iii).

Finally, let $\mathbf{c} = (y_n, \tau_n)$ be as in (iv), and set

$$u_n^{\mathbf{c}}(y, s) = u_n(y + y_n, s + \tau_n), \quad p_n^{\mathbf{c}}(y, s) = p_n(y + y_n, s + \tau_n).$$

The local-window condition, together with $\tau_n \leq 0$, is exactly what is needed to repeat the preceding estimates on each fixed backward compact cylinder in the (y, s) -variables: the physical cylinders remain in $B_\rho(x_*) \times (T - \rho^2, T)$, the local Type I envelope gives the same L^∞ bound on compact windows, and the pressure decomposition has the same local and finite-energy tail estimates. Thus the two failure modes in Theorem 8.3, blow-up of the local dissipation measure or failure of local pressure-gauge control, cannot occur for such a retained local recentering sequence after subsequence extraction. The recentered fields therefore converge locally to a bounded suitable terminal profile. If the local-window condition fails, then the recentering leaves every bounded terminal window attached to the singular point, or its critical mass persists only through expanding regions; those are precisely the exterior and diffuse terminal alternatives rather than loss of local compactness inside the local Type I bridge. $\square$

Lemma 1.11 (Local Type I concentration enters the terminal dichotomy). *Assume the realized critical-tail rigidity input Theorem 8.21. Under the hypotheses of Theorem 1.10, there exist $\eta_v > 0$ and a positive local velocity concentration sequence $r_n \downarrow 0$ at z_* such that*

$$(1.9) \quad \iint_{Q_1(0,0)} |u_n|^3 dx dt \geq \eta_v \quad \text{for all } n.$$

For the associated terminal extraction sequence (V_N, P_N) generated by a Seregin limit of this sequence, choose the small-data threshold $\varepsilon_{\text{sd}} \leq \eta_v$. Then the retained compact concentration generated from the origin recentering has no loss of local compactness. Any retained local recentering which stays in the local Type I window has the same compactness property by Theorem 1.10; any concentration leaving that window is routed to the exterior or diffuse terminal remainder. The paired terminal decomposition of Theorem 8.9, together with the mixed-state and active branch exclusions Theorems 8.22 and 9.16, reduces the retained possibilities to a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles.

Proof. By the local velocity concentration theorem of Paper 0, every singular point admits $\eta_v > 0$ and $r_n \downarrow 0$ such that

$$r_n^{-2} \iint_{B_{r_n}(x_*) \times (T - r_n^2, T)} |u|^3 dx dt \geq \eta_v.$$

After the rescaling this is exactly (1.9). By Theorem 1.10, the rescaled sequence has local Seregin compactness on every compact terminal cylinder, and therefore the origin concentration produces a bounded ancient Seregin profile with retained local L^3 -mass.

The local CKN measure μ_N in Theorem 8.3 dominates $|V_N|^3 dy d\tau$. Choosing $\varepsilon_{sd} \leq \eta_v$ therefore makes the distinguished origin recentering ε_{sd} -concentrating. The origin recentering satisfies the local-window condition in Theorem 1.10, so it is not in the local-compactness-failure class. Other retained local recenterings inside the same Type I window have the same compactness. Recenterings leaving the window are exterior or diffuse terminal remainders. The paired decomposition Theorem 8.9 records these remainders alongside the active profile locus rather than treating them as mutually exclusive alternatives. The sole-carrier radiative cases are excluded by Theorem 8.12, mixed compact–diffuse or compact–noncompact coexistence is excluded by Theorem 8.22, and infinite active branches are excluded by Theorem 9.16. Hence Theorem 8.10 leaves only a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles. $\square$

Theorem 1.12 (Local Type I singularities enter the full exclusion pipeline). *Assume the realized critical-tail rigidity input Theorem 8.21. Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that z_* satisfies the local Type I bound (1.6). Then there exists a bounded centered Type I Seregin profile with retained local concentration generated from (u, p, z_*) . Consequently the local terminal concentration package of Theorem 9.20 applies. In particular, no finite-energy local Type I singular point exists.*

Proof. Apply Theorem 1.11. The local Type I rescalings therefore produce at least one bounded terminal profile carrying retained local concentration. After passing to the centered variables, the bound $|U(x, t)| \leq M(-t)^{-1/2}$ becomes the global centered Type I bound

$$|V(y, \tau)| \leq M,$$

so each retained profile generated by the bridge is a bounded centered Type I Seregin profile.

If every such retained profile belongs to one of the previously closed lower classes, then the contradiction is supplied by the earlier papers: small-amplitude and stationary L^3 profiles are excluded in Paper I, the uniformly tight class is excluded in Paper II, and the closed structured/decay class is excluded in Paper III. Otherwise choose a retained profile outside those lower classes. It is then a refined residual candidate, so Theorem 9.20 applies. In that residual setting, a finite separated family of retained concentrating terminal profiles is excluded by Theorem 9.19, infinite active branches are excluded by Theorem 9.16, and mixed compact–diffuse tails are excluded by Theorem 8.22. The remaining single retained profile is terminally indecomposable, has the backward finite-critical-norm sequence, has parasitic-free finite-shift mildness, and is then excluded by the Albritton–Barker endpoint Liouville input used in this paper. Both routes are impossible. Hence the assumed finite-energy local Type I singular point does not exist. $\square$

Corollary 1.13 (Local Type I exclusion under realized critical-tail rigidity). *Assume the realized critical-tail rigidity input Theorem 8.21. Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$. If $z_* = (x_*, T)$ is a singular point, then z_* cannot satisfy the local Type I bound (1.6). Equivalently, in the local Type I/local Type II dichotomy of Paper 0, every finite-energy singular point belongs to the local Type II alternative.*

Proof. The first assertion is exactly Theorem 1.12. The equivalence is the local Type I/local Type II dichotomy of Paper 0: at a singular point, either some local Type I bound of the form (1.6) holds, or every such bound fails. $\square$

2. SHARED LOCAL NOTATION

For $z_0 = (y_0, \tau_0)$ and $r > 0$, write

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0).$$

For a centered profile (V, P) , the pressure-normalized local critical quantity is

$$(2.1) \quad \mathcal{E}_r(V, P; z_0) := \iint_{Q_r(z_0)} \left(|V|^3 + |P - c_{r,z_0}(\tau)|^{3/2} \right) dy d\tau,$$

where $c_{r,z_0}(\tau)$ is an allowed time-dependent pressure gauge, usually chosen as a spatial average over $B_r(y_0)$. All local pressure convergences below are understood after such gauges.

Definition 2.1 (Retained local concentration profile). Fix $\varepsilon_* > 0$. A terminal profile (U, Π) has retained local concentration if, after the terminal normalization under consideration,

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Definition 2.2 (Threshold ledger). The constants used in the terminal package are chosen in the following order. The compact velocity lower bounds η_0 and η_v are supplied by Paper I and Paper 0, respectively. The local small-data threshold ε_{sd} is then fixed with

$$0 < \varepsilon_{\text{sd}} \leq \min\{\eta_0, \eta_v, \varepsilon_{\text{CKN}}\},$$

where ε_{CKN} denotes the universal CKN regularity threshold whenever that threshold is invoked. The retained descendant threshold ε_* is chosen after the pressure gauges and compactness topology are fixed, with

$$0 < \varepsilon_* \leq \frac{1}{2} \varepsilon_{\text{sd}}.$$

Whenever a compactness or lower-semicontinuity passage loses a harmless fixed factor, ε_* is replaced once and for all by a smaller positive value. No later argument chooses a new concentration threshold depending on a previously extracted profile tree.

Definition 2.3 (Refined residual candidate). A refined residual candidate is a bounded normalized Seregin ancient limit obtained from the Paper I extraction theorem which carries the positive compact local velocity mass in (1.5) and which is not in any alternative already closed before the refined residual decomposition is entered. The definition is local: the only persistent lower bound attached to the candidate by Paper I is the fixed compact velocity L^3 lower bound.

Remark 2.4 (No hidden global L^3 assertion). The terminal arguments below do not claim that a generic residual profile is globally L^3 -tight or globally bounded in $L^\infty_\tau L^3_y$. The only nonlocal critical-norm conclusion used at the endpoint is the weaker sequence- L^3 completeness of retained indecomposable concentration profiles, and that conclusion is derived from local terminal alternatives rather than from an a priori global estimate on the original residual class.

3. EXCLUSION OF THE AXISYMMETRIC BOUNDED-CIRCULATION RESIDUAL CLASS

In this section we prove the rigidity statement needed for the first refined residual class. The argument uses the scale-invariant angular circulation in centered variables. The essential point is that the centered Type I equation contains the Ornstein–Uhlenbeck drift. Thus the angular-circulation equation has an absorbing boundary at the symmetry axis together with a confining drift at spatial infinity. This gives a scalar Liouville theorem for bounded ancient solutions of the circulation equation.

Throughout the section we write cylindrical coordinates as (ρ, θ, Z) , and an axisymmetric vector field as

$$V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The centered Navier–Stokes equation is

$$(3.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For an axisymmetric solution define the centered angular circulation

$$(3.2) \quad \Gamma(\rho, Z, \tau) := \rho V_\theta(\rho, Z, \tau).$$

This is the same scale-invariant quantity as the physical circulation rU_θ . If the physical representative is

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)),$$

then

$$rU_\theta(r, z, t) = \rho V_\theta(\rho, Z, \tau), \quad r = \sqrt{-t} \rho, \quad z = \sqrt{-t} Z.$$

Lemma 3.1 (Centered circulation equation). *Let (V, P) be a smooth bounded axisymmetric solution of (3.1). Then $\Gamma = \rho V_\theta$ satisfies*

$$(3.3) \quad \partial_\tau \Gamma = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma - \left(V_\rho + \frac{\rho}{2}\right) \partial_\rho \Gamma - \left(V_Z + \frac{Z}{2}\right) \partial_Z \Gamma$$

on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Moreover

$$(3.4) \quad \Gamma(0, Z, \tau) = 0 \quad \text{for all } (Z, \tau) \in \mathbb{R} \times \mathbb{R}.$$

Proof. The angular component of (3.1) is

$$\partial_\tau V_\theta - \left(\Delta - \frac{1}{\rho^2}\right) V_\theta + \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta + V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} = 0.$$

Multiplying by ρ , using

$$\begin{aligned} \rho \left(\Delta - \frac{1}{\rho^2}\right) V_\theta &= \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma, \\ \rho \left[\frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta\right] &= \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) \Gamma, \end{aligned}$$

and

$$\rho \left(V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho}\right) = V_\rho \partial_\rho \Gamma + V_Z \partial_Z \Gamma,$$

gives (3.3). Finally, smoothness of a bounded axisymmetric vector field at the axis implies $V_\theta(\rho, Z, \tau) = O(\rho)$ as $\rho \downarrow 0$, locally uniformly in (Z, τ) . Hence $\Gamma = \rho V_\theta = O(\rho^2)$, which gives (3.4). $\square$

We next record the scalar Liouville theorem used below. It is stated in a form that isolates precisely the PDE mechanism: bounded drift perturbations of the centered circulation operator cannot sustain a bounded ancient solution which vanishes on the symmetry axis.

Lemma 3.2 (Axis-absorption Liouville theorem). *Let $B = (B_\rho, B_Z)$ be a smooth vector field on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$ satisfying*

$$\|B\|_{L^\infty} < \infty.$$

Let $G \in C^{2,1}((0, \infty) \times \mathbb{R} \times \mathbb{R}) \cap L^\infty$ solve

$$(3.5) \quad \partial_\tau G = \partial_\rho^2 G - \frac{1}{\rho} \partial_\rho G + \partial_Z^2 G - \left(B_\rho + \frac{\rho}{2}\right) \partial_\rho G - \left(B_Z + \frac{Z}{2}\right) \partial_Z G$$

for all $\tau \in \mathbb{R}$, and assume

$$(3.6) \quad \lim_{\rho \downarrow 0} G(\rho, Z, \tau) = 0 \quad \text{locally uniformly in } (Z, \tau).$$

Then $G \equiv 0$.

Proof. Set $M_B = \|B\|_{L^\infty}$. For $R > 1$ define

$$(3.7) \quad h_R(\rho) := \frac{\int_0^{\min\{\rho, R\}} s \exp\left(\frac{s^2}{4} - M_B s\right) ds}{\int_0^R s \exp\left(\frac{s^2}{4} - M_B s\right) ds}.$$

Then $0 \leq h_R \leq 1$, $h_R(0) = 0$, $h_R(R) = 1$, and, on $0 < \rho < R$,

$$(3.8) \quad h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0, \quad h_R' > 0.$$

Let

$$\mathcal{L}_\tau f := \partial_\rho^2 f - \frac{1}{\rho} \partial_\rho f + \partial_Z^2 f - \left(B_\rho + \frac{\rho}{2}\right) \partial_\rho f - \left(B_Z + \frac{Z}{2}\right) \partial_Z f.$$

Since $B_\rho \geq -M_B$, (3.8) gives

$$(3.9) \quad \mathcal{L}_\tau h_R = h_R'' - \left(\frac{1}{\rho} + \frac{\rho}{2} + B_\rho\right) h_R' \leq h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0.$$

Thus $(\partial_\tau - \mathcal{L}_\tau)h_R \geq 0$.

We now compare G to this barrier on the strip $\Omega_R = (0, R) \times \mathbb{R}$. Let $M_G = \|G\|_{L^\infty}$. We first record the killed-strip decay estimate used by the comparison argument. If $q_{s,R}$ solves

$$\partial_\tau q_{s,R} = \mathcal{L}_\tau q_{s,R}, \quad q_{s,R} = 0 \text{ on } \rho = 0, R, \quad q_{s,R}(\rho, Z, s) = 1,$$

then, for every compact $K \Subset \Omega_R$ and every fixed t ,

$$(3.10) \quad \lim_{s \rightarrow -\infty} \sup_{(\rho, Z) \in K} q_{s,R}(\rho, Z, t) = 0.$$

To prove (3.10), fix $K \Subset \Omega_R$. Choose $\delta > 0$, $Z_K > 0$, and $Z_1 > Z_K + 1$ so that

$$K \subset [\delta, R - \delta] \times [-Z_K, Z_K] \subset (0, R) \times (-Z_1, Z_1).$$

On the finite cylinder

$$\Omega_{R,Z_1} := (0, R) \times (-Z_1, Z_1)$$

with Dirichlet data on the full parabolic boundary, the operator $\mathcal{L}_\tau$ is uniformly parabolic on $[\delta/2, R - \delta/2] \times [-Z_1, Z_1]$, has bounded first-order coefficients there, and has absorbing boundary at $\rho = 0, R$. The boundary parabolic Harnack inequality and the strong maximum principle give a time $T_R > 0$ and a number $\vartheta_R \in (0, 1)$, depending only on R, M_B, δ, Z_K and not on Z_1 , such that every nonnegative solution with initial data bounded by 1 and zero lateral data satisfies

$$0 \leq q(\cdot, s + T_R) \leq 1 - \vartheta_R \quad \text{on } [\delta, R - \delta] \times [-Z_K, Z_K].$$

Applying the same estimate after each time translation and using the maximum principle between the compact set and the boundary gives

$$\sup_K q_{s,R}(\cdot, s + mT_R) \leq (1 - \vartheta_R)^m, \quad m = 1, 2, \dots$$

For a fixed $t > s$, choose m with $s + mT_R \leq t < s + (m+1)T_R$. The maximum principle and interior Harnack estimate on the final interval $[s + mT_R, t]$ preserve the preceding bound up to a constant depending only on R, M_B, δ, Z_K, T_R . Letting $s \rightarrow -\infty$ forces $m \rightarrow \infty$, proving decay on K for the truncated problem. The solutions on Ω_{R,Z_1} increase monotonically to the killed-strip solution as $Z_1 \rightarrow \infty$; passing to this limit gives (3.10).

Fix $R > 1$, $s < t$, and set

$$W_R^+ := \frac{G}{M_G} - h_R, \quad W_R^- := -\frac{G}{M_G} - h_R,$$

with the convention that the conclusion is immediate if $M_G = 0$. By (3.5) and (3.9),

$$(\partial_\tau - \mathcal{L}_\tau)W_R^\pm \leq 0 \quad \text{in } \Omega_R \times (s, t).$$

On $\rho = 0$, (3.6) and $h_R(0) = 0$ give $W_R^\pm \leq 0$. On $\rho = R$, $h_R(R) = 1$ and $|G| \leq M_G$ give again $W_R^\pm \leq 0$. The parabolic comparison principle with the killed-strip kernel therefore yields, for $\tau = t$,

$$(W_R^\pm)_+(\rho, Z, t) \leq q_{s,R}(\rho, Z, t) \quad ((\rho, Z) \in \Omega_R).$$

Letting $s \rightarrow -\infty$ and using (3.10) gives $W_R^\pm \leq 0$ on compact subsets of Ω_R . Since the compact subset is arbitrary,

$$(3.11) \quad |G(\rho, Z, t)| \leq M_G h_R(\rho) \quad (0 < \rho < R, Z \in \mathbb{R}, t \in \mathbb{R}).$$

Finally, for each fixed $\rho < \infty$, the denominator in (3.7) tends to infinity super-exponentially as $R \rightarrow \infty$, while the numerator remains fixed. Hence $h_R(\rho) \rightarrow 0$. Letting $R \rightarrow \infty$ in (3.11) gives $G(\rho, Z, t) = 0$ for every $\rho > 0$, $Z \in \mathbb{R}$, and $t \in \mathbb{R}$. The boundary value at $\rho = 0$ is already zero by assumption. Therefore $G \equiv 0$. $\square$

Theorem 3.3 (Exclusion of the axisymmetric bounded-circulation residual class). *Let $\mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$ be the axisymmetric bounded-circulation residual class in the refined residual decomposition. Then*

$$(3.12) \quad \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Assume, for contradiction, that $V \in \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$. By definition of $\mathcal{C}_{\text{ax}}$, the associated physical ancient representative is axisymmetric and has bounded angular circulation. Equivalently, in centered variables,

$$\Gamma(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau)$$

is bounded on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Since V is a Seregin ancient limit, it is smooth and bounded in centered variables. Hence $B = (V_\rho, V_Z)$ is bounded.

By Theorem 3.1, Γ satisfies (3.3) and vanishes at the axis. Applying Theorem 3.2 with $G = \Gamma$ and $B = (V_\rho, V_Z)$ gives

$$\Gamma \equiv 0.$$

Therefore $V_\theta \equiv 0$ for $\rho > 0$, and by smoothness also on the axis. Thus the associated ancient solution is axisymmetric without swirl. But the axisymmetric no-swirl class is one of the structured alternatives removed before the residual class is defined. This contradicts $V \in \mathcal{R}(\mathcal{S}; I, J)$, and therefore no such V exists. $\square$

Remark 3.4 (Why this does not assert the full bounded-swirl Liouville theorem). The proof uses the centered Type I equation, specifically the Ornstein–Uhlenbeck drift terms $-\rho\partial_\rho/2$ and $-Z\partial_Z/2$ in the scalar equation for Γ . It is therefore a Type I Seregin limit argument. It does not prove that every bounded ancient axisymmetric solution of the unrenormalized Navier–Stokes equations with bounded physical circulation is swirl-free. The latter lacks the confining centered drift and remains a different shell-transport problem.

4. THE ROTATIONAL RELATIVE-EQUILIBRIUM RESIDUAL CLASS

We next isolate the second refined residual class. This class consists of centered Type I ancient limits whose time dependence is a rigid rotation of a fixed spatial profile. Passing to the co-rotating frame converts the centered Navier–Stokes equation into a stationary Leray profile equation with a Coriolis transport term. A whole-space Coriolis–Leray Liouville theorem for all bounded profiles is not used here. Instead, the Coriolis term is localized as an annular flux and is then treated by the local concentration analysis proved in Sections 8 and 9.

Throughout this section the centered equation is

$$(4.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For $\Omega \in \mathfrak{so}(3)$, set

$$(4.2) \quad Q(\tau) := \exp(\tau\Omega) \in SO(3).$$

Definition 4.1 (Rotational relative-equilibrium profile). A bounded smooth ancient solution (V, P) of (4.1) is a rotational relative equilibrium if there exist $\Omega \in \mathfrak{so}(3)$, a bounded smooth divergence-free vector field W , and a pressure profile Π , such that

$$(4.3) \quad V(y, \tau) = Q(\tau)W(Q(\tau)^T y), \quad P(y, \tau) = \Pi(Q(\tau)^T y)$$

up to the usual time-dependent pressure gauge. The class $\mathcal{C}_{\text{rot}}$ consists of such residual Seregin profiles, after the axisymmetric bounded-circulation class and the previously closed Paper I–III alternatives have been removed.

Lemma 4.2 (Co-rotating stationary equation). *Let (V, P) be a rotational relative equilibrium. Then the co-rotating profile (W, Π) satisfies*

$$(4.4) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Equivalently,

$$(4.5) \quad \mathbb{P} \left[-\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W \right] = 0,$$

where $\mathbb{P}$ is the whole-space Leray projector.

Proof. Let $z = Q(\tau)^T y$. Since $Q'(\tau) = \Omega Q(\tau)$ and $Q(\tau)^T \Omega Q(\tau) = \Omega$, one has

$$\partial_\tau z = -\Omega z.$$

Therefore

$$\partial_\tau V(y, \tau) = Q(\tau) \left(\Omega W(z) - (\Omega z) \cdot \nabla W(z) \right).$$

The Laplacian, divergence, convection term, and pressure gradient are invariant under the orthogonal change of variables. Substitution into (4.1), followed by multiplication by $Q(\tau)^T$, gives (4.4). Applying the Leray projector gives (4.5). $\square$

4.1. The Bernoulli identity and the local Coriolis flux. Write

$$(4.6) \quad F(y) := \frac{1}{2}y - \Omega y, \quad A(y) := W(y) + F(y), \quad \omega := \nabla \times W.$$

Since Ω is skew-symmetric, there is a unique vector $\varpi \in \mathbb{R}^3$ such that

$$(4.7) \quad \Omega y = \varpi \times y.$$

Then

$$(4.8) \quad \nabla \times F = -2\varpi.$$

Lemma 4.3 (Coriolis Bernoulli identity). *Let (W, Π) be a smooth solution of (4.4). Define*

$$(4.9) \quad B := \frac{1}{2}|W|^2 + \Pi + F \cdot W.$$

Then

$$(4.10) \quad \nabla B + \nabla \times \omega - A \times \omega = 0,$$

and consequently

$$(4.11) \quad -\Delta B + A \cdot \nabla B = -|\omega|^2 - \omega \cdot (\nabla \times F) = -|\omega|^2 + 2\varpi \cdot \omega.$$

Proof. The vector identity

$$(W \cdot \nabla)W = \nabla \left(\frac{1}{2}|W|^2 \right) - W \times \omega$$

and the identity

$$(F \cdot \nabla)W + (\nabla F)^T W = \nabla(F \cdot W) - F \times \omega$$

apply because $(\nabla F)^T W = \frac{1}{2}W + \Omega W$. Thus (4.4) can be rewritten as

$$-\Delta W + \nabla \left(\frac{1}{2}|W|^2 + \Pi + F \cdot W \right) - (W + F) \times \omega = 0.$$

Since $\nabla \cdot W = 0$, one has $-\Delta W = \nabla \times \omega$, which gives (4.10).

Taking divergence in (4.10) gives

$$\Delta B = \nabla \cdot (A \times \omega),$$

because the divergence of a curl vanishes. Using

$$\nabla \cdot (A \times \omega) = \omega \cdot (\nabla \times A) - A \cdot (\nabla \times \omega), \quad \nabla \times A = \omega + \nabla \times F,$$

we obtain

$$\Delta B = |\omega|^2 + \omega \cdot (\nabla \times F) - A \cdot (\nabla \times \omega).$$

On the other hand, by (4.10),

$$A \cdot \nabla B = A \cdot (A \times \omega) - A \cdot (\nabla \times \omega) = -A \cdot (\nabla \times \omega).$$

Combining the last two identities gives (4.11). Finally, (4.8) follows from (4.7). $\square$

Lemma 4.4 (Localized Coriolis flux identity). *Let (W, Π) be as in Theorem 4.3, and let $\chi \in C_c^\infty(\mathbb{R}^3)$. Then*

$$(4.12) \quad \int_{\mathbb{R}^3} \chi (-\Delta B + A \cdot \nabla B + |\omega|^2) dy = -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy.$$

In particular the Coriolis contribution is supported only where the cutoff varies.

Proof. By (4.11),

$$-\Delta B + A \cdot \nabla B + |\omega|^2 = 2\varpi \cdot \omega.$$

Because ϖ is constant and $\omega = \nabla \times W$,

$$\varpi \cdot \omega = \nabla \cdot (W \times \varpi).$$

Multiplying by χ , integrating over $\mathbb{R}^3$, and integrating by parts gives

$$2 \int \chi \varpi \cdot \omega = 2 \int \chi \nabla \cdot (W \times \varpi) = -2 \int (W \times \varpi) \cdot \nabla \chi,$$

which proves (4.12). $\square$

Lemma 4.5 (Non-small Coriolis flux creates a retained local concentration). *Let $\chi \in C_c^\infty(\mathbb{R}^3)$, let $A_\chi := \text{supp } \nabla \chi$, and put*

$$C_\chi := \int_{A_\chi} |\nabla \chi|^{3/2} dy.$$

Fix $T \in (0, 1]$, and let

$$\mathcal{T}_{\chi, T} := \{(y, \tau) : -T < \tau < 0, Q(\tau)^T y \in A_\chi\}$$

be the swept annular tube in the original centered variables. Cover $\mathcal{T}_{\chi,T}$ by finitely many unit terminal cylinders

$$\mathcal{T}_{\chi,T} \subset \bigcup_{m=1}^{N_{\chi,T}} Q_1(z_m).$$

Let $\varepsilon_{\text{sd}} > 0$ be the local concentration threshold used in [Theorem 8.3](#). If $\varpi \neq 0$ and

$$(4.13) \quad \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right| \geq 2|\varpi| C_{\chi}^{2/3} \left(\frac{N_{\chi,T} \varepsilon_{\text{sd}}}{T} \right)^{1/3},$$

then at least one cylinder $Q_1(z_m)$ is ε_{sd} -concentrating for the original centered solution V .

Proof. Write

$$\delta_{\chi} := \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right|.$$

Holder's inequality gives

$$\begin{aligned} \delta_{\chi} &\leq 2|\varpi| \int_{A_{\chi}} |W| |\nabla \chi| \, dy \\ &\leq 2|\varpi| \left(\int_{A_{\chi}} |W|^3 \, dy \right)^{1/3} C_{\chi}^{2/3}. \end{aligned}$$

Thus (4.13) implies

$$\int_{A_{\chi}} |W|^3 \, dy \geq \frac{N_{\chi,T} \varepsilon_{\text{sd}}}{T}.$$

Since $V(y, \tau) = Q(\tau)W(Q(\tau)^T y)$ and $Q(\tau)$ is an isometry,

$$\begin{aligned} \iint_{\mathcal{T}_{\chi,T}} |V(y, \tau)|^3 \, dy \, d\tau &= \int_{-T}^0 \int_{Q(\tau)A_{\chi}} |W(Q(\tau)^T y)|^3 \, dy \, d\tau \\ &= T \int_{A_{\chi}} |W(z)|^3 \, dz \geq N_{\chi,T} \varepsilon_{\text{sd}}. \end{aligned}$$

Because the tube is covered by $N_{\chi,T}$ unit terminal cylinders, one of them satisfies

$$\iint_{Q_1(z_m)} |V|^3 \, dy \, d\tau \geq \varepsilon_{\text{sd}}.$$

The pressure contribution to the local CKN measure is nonnegative after the allowed pressure gauge. Hence

$$\mu(Q_1(z_m)) \geq \varepsilon_{\text{sd}},$$

so the corresponding recentering sequence is ε_{sd} -concentrating in the sense of [Theorem 8.3](#). $\square$

Lemma 4.6 (Local Coriolis-flux dichotomy). *Fix $\chi \in C_c^{\infty}(\mathbb{R}^3)$ and $T \in (0, 1]$. Along every terminal extraction of a rotational relative equilibrium, exactly one of the following local alternatives occurs:*

- (i) $\varpi \neq 0$, the flux across $A_{\chi} = \text{supp } \nabla \chi$ satisfies the threshold (4.13), and the swept annular tube contains a unit-scale concentration;
- (ii) either $\varpi = 0$, or the threshold (4.13) fails; then the flux is a bounded compactly supported functional of W on A_{χ} , passes to terminal limits by local L^3 convergence, and creates no terminal alternative other than the single-profile, finite separated-profile, exterior-escape, diffuse-exterior-concentration, local-vanishing, or local-compactness-failure alternatives of [Theorem 8.9](#).

Proof. Alternative (i) is precisely [Theorem 4.5](#). If $\varpi = 0$, then the flux vanishes identically. If $\varpi \neq 0$ and (4.13) fails, the flux is bounded by the right hand side of that threshold. In both cases the functional

$$W \mapsto -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy$$

depends only on W on the compact annular set A_χ . Under the local Seregin compactness supplied by Paper I, velocities converge strongly in L^3_{loc} ; since $\nabla \chi \in L^{3/2}$ and ϖ is fixed, the functional is continuous along every terminal subsequence:

$$\int (W_n \times \varpi) \cdot \nabla \chi \longrightarrow \int (W_\infty \times \varpi) \cdot \nabla \chi.$$

Thus a subthreshold flux is retained only as a local term in the limiting localized Bernoulli identity (4.12). It is not a new location of local velocity mass. Every actual location of compact velocity L^3 -mass is still one of the terminal local concentration alternatives listed in [Theorem 8.9](#). $\square$

Remark 4.7 (Why this is not a global Liouville argument). For $\Omega = 0$, the Coriolis flux in [Theorem 4.4](#) vanishes. For $\Omega \neq 0$, the same lemma does not give a sign-definite maximum principle for B . It gives only the local dichotomy in [Theorem 4.6](#): either a chosen annulus already contains a recentered local concentration, or the annular flux is a subthreshold compact functional which passes to the local terminal limit. No tail decay, whole-space L^p bound, Gaussian energy estimate, or global pressure normalization is used.

4.2. Local closure of the rotational relative-equilibrium class.

Theorem 4.8 (Stratified exclusion of rotational relative equilibria). *The ordered rotational relative-equilibrium residual class is empty:*

$$(4.14) \quad \mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{rot}}(\mathcal{S}; I, J)$. By [Theorem 4.1](#) and [Theorem 4.2](#), there are $\Omega \in \mathfrak{so}(3)$ and a bounded smooth divergence-free profile (W, Π) satisfying the Coriolis–Leray equation (4.4). The centered representative V is a normalized Seregin ancient limit, so by [Theorem 1.3](#) it carries a fixed positive compact velocity L^3 -mass.

Choose a countable family of smooth cutoffs which is cofinal among compact annuli in the terminal variables. For each cutoff and each rational $T \in (0, 1]$, apply the local dichotomy [Theorem 4.6](#). A superthreshold Coriolis flux produces a unit-scale concentration and is therefore already part of the local concentration decomposition. A subthreshold Coriolis flux is a compactly supported functional of W , continuous under the local convergence supplied by Paper I, and hence remains inside the localized limiting equation; it carries no independent CKN mass and creates no additional terminal alternative.

We now apply the terminal local concentration decomposition to the inherited compact density. Loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration cannot be the sole locations of this density by [Theorem 8.12](#). By the preceding paragraph, the Coriolis flux has already led either to a local concentration or to a compact local term in the selected terminal equation. Thus, after local concentration extraction, the retained density is located either in a finite separated family of retained concentration profiles or in a single retained concentration terminal profile.

The finite separated-family alternative is impossible by [Theorem 9.19](#). In the single-profile case, [Theorem 9.4](#) makes the terminal profile terminally indecomposable, and [Theorem 9.8](#) excludes retained indecomposable concentration profiles. Both possible locations of the inherited compact density are impossible. This contradicts the normalization of V , and therefore $\mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset$. $\square$

Proposition 4.9 (Already-closed integrable subcase). *Let V be a rotational relative equilibrium with co-rotating profile W . If*

$$(4.15) \quad W \in L^3(\mathbb{R}^3),$$

then $V \equiv 0$. More generally, the same conclusion holds under any condition which places the physical representative in $L_t^\infty L_x^3$ on its ancient time interval.

Proof. Rotations and the Navier–Stokes critical scaling preserve the L^3 norm. Thus $W \in L^3$ implies

$$\sup_{\tau \in \mathbb{R}} \|V(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Equivalently, the physical representative lies in the critical class $L_t^\infty L_x^3$. The endpoint L^3 class is one of the already closed alternatives in Paper I, by the Escauriaza–Seregin–Šverák endpoint regularity theorem [7] and the stationary Nečas–Růžička–Šverák Liouville theorem [11]. Therefore $V \equiv 0$. A zero profile has zero local velocity mass, so no such profile can belong to the residual nonvanishing class. $\square$

5. THE DEGENERATE STATIONARY-HULL RESIDUAL CLASS

We now isolate the third refined residual class. This class is designed to capture Type I ancient limits whose time-translation hull contains a stationary profile with retained local concentration lying on a non-isolated family of stationary centered profiles. The role of the stationary-family condition is to separate genuinely degenerate stationary profile families from isolated stationary profiles and from the symmetry-generated classes already removed in $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$.

A small point of formulation is important. It is not enough to assume that the hull of a residual ancient solution contains some stationary-family element. A stationary element may occur as a limit without retained local concentration and therefore need not retain the local CKN concentration inherited from the original singular point. Such a hull element should not be placed in a class that is to be closed by a stationary-profile exclusion. Therefore the class below is defined using *retained-concentration occurrence*: the stationary-family profile must itself occur as a Seregin limit carrying a positive local CKN concentration.

Throughout this section the centered equation is

$$(5.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

Its projected form is

$$(5.2) \quad \partial_\tau V = \mathcal{F}(V), \quad \mathcal{F}(W) := \mathbb{P} \left[\Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W \right],$$

where $\mathbb{P}$ is the whole-space Leray projector.

5.1. Stationary families and retained-concentration hull limits.

Definition 5.1 (Admissible stationary phase space). An admissible stationary phase space is a Banach space $\mathfrak{B}$ of solenoidal vector fields on $\mathbb{R}^3$ such that:

- (i) $\mathfrak{B} \hookrightarrow C_{\text{loc}}^{2,\alpha}(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$ continuously for some $\alpha \in (0, 1)$;
- (ii) the map $\mathcal{F}$ in (5.2) is well-defined on an open subset $\mathcal{U} \subset \mathfrak{B}$ and maps $\mathcal{U}$ into a Banach space $\mathfrak{V}$;
- (iii) $\mathcal{F} : \mathcal{U} \rightarrow \mathfrak{V}$ is C^1 .

Definition 5.2 (Nontrivial stationary family). Let $W \in \mathcal{U} \subset \mathfrak{B}$ satisfy $\mathcal{F}(W) = 0$. We say that W lies on an admissible nontrivial stationary family if there exist $\varepsilon > 0$ and a C^1 curve

$$a \mapsto W_a \in \mathcal{U}, \quad |a| < \varepsilon,$$

such that

$$(5.3) \quad W_0 = W, \quad \mathcal{F}(W_a) = 0 \quad (|a| < \varepsilon),$$

and such that the family is not locally generated only by the action of the rigid rotation group. Equivalently, after possibly reparametrizing the curve, one may assume that

$$(5.4) \quad \dot{W}_0 \notin \{\Omega W - (\Omega y) \cdot \nabla W : \Omega \in \mathfrak{so}(3)\}$$

whenever the tangent $\dot{W}_0$ exists and is nonzero.

Definition 5.3 (Retained local CKN concentration). Let (W, Π) be a stationary centered profile. We say that (W, Π) is ε_* -concentrating if there exist a compact parabolic cylinder $Q_r(z_0)$ and an admissible pressure gauge c_{r,z_0} such that

$$(5.5) \quad \iint_{Q_r(z_0)} (|W|^3 + |\Pi - c_{r,z_0}|^{3/2}) \, dy \, d\tau \geq \varepsilon_*.$$

For a stationary profile the integrand is independent of τ , but the parabolic-cylinder notation is retained to match the CKN normalization used for Seregin limits.

Definition 5.4 (Degenerate stationary-hull residual class with retained concentration). Fix an admissible stationary phase space $\mathfrak{B}$. The degenerate stationary-hull residual class with retained local concentration

$$\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$$

consists of those $V \in \mathcal{R}(\mathcal{S}; I, J)$, after the ordered removal of $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$, for which there exists a sequence $s_k \in \mathbb{R}$ and a stationary profile (W, Π) such that

$$(5.6) \quad (V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \longrightarrow (W, \Pi)$$

locally smoothly for the velocity and locally in $L^{3/2}$ for the pressure modulo admissible gauges, and such that:

- (i) $W \in \mathfrak{B}$ and $\mathcal{F}(W) = 0$;
- (ii) W lies on an admissible nontrivial stationary family in the sense of [Theorem 5.2](#);
- (iii) (W, Π) is ε_* -concentrating in the sense of [Theorem 5.3](#);
- (iv) (W, Π) is not in any previously closed stationary, structured, axisymmetric bounded-circulation, or rotational relative-equilibrium class.

In the ordered residual decomposition one keeps this class of stationary-hull limits with retained concentration as the degenerate stationary-hull alternative. The hull cases without retained local concentration should be assigned to the generic terminal residual class, where they are handled by the terminal local concentration decomposition rather than by a stationary-profile exclusion.

Remark 5.5 (Why retained concentration is necessary). The condition (5.5) prevents the following logical error. A nonstationary ancient solution may have a stationary profile in its hull with zero local CKN mass on the relevant compact cylinders. Excluding that stationary profile would not exclude the original ancient solution. The stationary-profile exclusion used below applies only to stationary profiles that themselves occur as nontrivial Seregin limits with positive local concentration.

5.2. Hull limits are again Seregin limits. The next lemma is the basic compatibility between time-translation hulls in centered variables and rescaling sequences in physical variables. It is the reason that a stationary profile in the hull which retains local concentration may be treated as a blow-up profile of the original solution, not merely as a limit inside the already extracted ancient solution.

Lemma 5.6 (Scale reselection for centered hull limits). *Let (V, P) be a centered Type I Seregin limit obtained from a finite-energy suitable weak solution (u, p) at a singular point $z_* = (x_*, t_*)$. Suppose that, for some sequence $s_k \in \mathbb{R}$,*

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \rightarrow (W, \Pi)$$

locally in the Seregin topology. Then, after passing to a diagonal subsequence and reselecting the physical blow-up scales, (W, Π) is itself a centered Type I Seregin limit of (u, p) at z_ . If the convergence to (W, Π) retains a positive local CKN lower bound, then (W, Π) is a nontrivial Seregin limit with the same type of inherited nonvanishing condition.*

Proof. Let $\lambda_n \downarrow 0$ be a sequence of scales producing (V, P) . In centered variables, a time translation by s is exactly a change of the physical blow-up scale by a factor depending only on s . With the convention used in (5.1), write this adjusted scale as

$$(5.7) \quad \lambda_{n,s} := e^{-s/2} \lambda_n.$$

The rescaled fields built from (u, p) at scale $\lambda_{n,s}$ are the same as the fields at scale λ_n , translated by s in the centered time variable, up to the corresponding pressure gauge.

Fix an exhaustion of compact cylinders $K_m \subset \mathbb{R}^3 \times \mathbb{R}$. For each k , choose $n(k)$ so large that $\lambda_{n(k), s_k} \rightarrow 0$ and the rescaled fields at scale $\lambda_{n(k)}$, shifted by s_k , are within $1/k$ of $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ on K_k in the local Seregin topology. This is possible because $\lambda_n \downarrow 0$ and because the original sequence converges locally to (V, P) .

By the assumed hull convergence, the shifted limit $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ converges to (W, Π) on each fixed compact cylinder. The diagonal sequence of physical rescalings with scales $\lambda_{n(k), s_k}$ therefore converges locally to (W, Π) . Hence (W, Π) is again a centered Type I Seregin limit at z_* . The assertion about nonvanishing follows from lower semicontinuity and local convergence of the CKN quantities after the prescribed pressure gauges. $\square$

5.3. Elementary consequences of stationary-family degeneracy. The stationary-family condition has two immediate PDE consequences. First, a differentiable nontrivial family produces a nontrivial element of the kernel of the linearized stationary Leray operator. Second, if an ancient trajectory is exactly constrained to a stationary family, then it is stationary.

Lemma 5.7 (Stationary family implies linearized degeneracy). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family satisfying (5.3). Then*

$$(5.8) \quad D\mathcal{F}(W_0)\dot{W}_0 = 0.$$

In particular, if $\dot{W}_0 \neq 0$ modulo infinitesimal rotations, the linearized stationary Leray operator has a genuine non-symmetry kernel element.

Proof. Differentiate $\mathcal{F}(W_a) = 0$ at $a = 0$. Since $\mathcal{F}$ is C^1 on $\mathfrak{B}$, this gives (5.8). The final statement is exactly the nontriviality condition in Theorem 5.2. $\square$

Lemma 5.8 (Trajectories constrained to a stationary family are stationary). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family with $\dot{W}_a \neq 0$ on an interval. Suppose that a smooth solution of (5.2) has the form*

$$(5.9) \quad V(\tau) = W_{a(\tau)}$$

for a C^1 function $a(\tau)$. Then a is constant, and hence V is stationary.

Proof. Since $\mathcal{F}(W_a) = 0$ for every a , equation (5.2) gives

$$a'(\tau)\dot{W}_{a(\tau)} = \partial_\tau V(\tau) = \mathcal{F}(W_{a(\tau)}) = 0.$$

Because $\dot{W}_{a(\tau)} \neq 0$, we obtain $a'(\tau) = 0$. Thus a is constant on the interval. $\square$

Corollary 5.9 (Exact stationary-family subcase). *If an element of the residual class is itself an exact trajectory on an admissible stationary family, then it is a stationary centered profile. Hence it is excluded whenever it satisfies the stationary L^3 Liouville criterion, or more generally whenever the stationary profile belongs to one of the previously closed stationary or structured classes.*

Proof. Let $V(\tau) = W_{a(\tau)}$ be such an exact stationary-family trajectory. By Theorem 5.8, $a'(\tau) = 0$ on every interval on which the stationary-family parametrization is nondegenerate. Hence a is constant and V is a single stationary centered profile. If that stationary profile satisfies the stationary L^3 Liouville criterion or lies in one of the previously closed structured or ordered lower residual classes, the corresponding prior exclusion removes it from the residual class. $\square$

5.4. Stationary-hull profiles which occur as Seregin limits. The remaining issue is not the internal dynamics along a stationary family; exact stationary-family motion is stationary by Theorem 5.8. The real issue is occurrence as a blow-up limit: can a degenerate stationary-family profile occur as a Seregin limit of a finite-energy suitable weak solution while retaining local concentration? The following statement is the precise Seregin limit occurrence statement required to close the degenerate stationary-hull class with retained local concentration. It is proved in the following section by the terminal local concentration theorem.

Theorem 5.10 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. Let (W, Π) be a stationary centered profile satisfying:*

- (i) $W \in \mathfrak{B}$, $\mathcal{F}(W) = 0$, and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) is ε_* -concentrating;
- (iv) (W, Π) is not in the small, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, or rotational relative-equilibrium classes already removed earlier in the stratification.

Then no such (W, Π) exists.

Proof. This is exactly Theorem 6.5, proved below from the local terminal stratification theorem. $\square$

Remark 5.11 (Interpretation of the Seregin limit occurrence theorem). Theorem 5.10 excludes occurrence as a Type I Seregin limit. It does not assert that degenerate stationary profiles cannot exist as elliptic solutions of the stationary centered equation. It asserts only that such profiles cannot occur as Type I blow-up limits with retained local concentration after all previously closed classes have been removed. This is the correct formulation for the residual problem: the elliptic family may exist, but it cannot be reached by the local blow-up procedure coming from a finite-energy suitable weak solution.

5.5. Closure of the degenerate stationary-hull concentration class.

Theorem 5.12 (Exclusion of the degenerate stationary-hull concentration class).

$$(5.10) \quad \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Consequently the ordered third residual class is empty, provided it is defined with the retained-concentration Seregin limit condition in [Theorem 5.4](#).

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By [Theorem 5.4](#), there is a stationary profile (W, Π) in the centered time-translation hull of (V, P) , and (W, Π) is ε_* -concentrating. The profile W belongs to $\mathfrak{B}$, satisfies $\mathcal{F}(W) = 0$, and lies on an admissible nontrivial stationary family. Moreover, by the ordered definition of $\mathcal{C}_{\text{stat}}^{\text{ret}}$, it is not in any of the previously closed lower classes.

By [Theorem 5.6](#), the hull profile (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point, after reselecting scales. Since local concentration is retained, (W, Π) satisfies every condition in [Theorem 5.10](#). This contradicts the Seregin limit occurrence theorem. Therefore no such V exists, proving (5.10). $\square$

Proposition 5.13 (Already-closed integrable stationary-family subcase). *Let (W, Π) be a stationary centered profile lying on an admissible stationary family. If*

$$(5.11) \quad W \in L^3(\mathbb{R}^3),$$

then $W \equiv 0$. Hence no retained-concentration hull profile satisfying (5.11) can occur in $\mathcal{C}_{\text{stat}}^{\text{ret}}$.

Proof. The stationary physical representative associated with W is a backward self-similar Navier–Stokes solution. The assumption $W \in L^3(\mathbb{R}^3)$ places it in the stationary L^3 class already excluded by the classical Nečas–Růžička–Šverák Liouville theorem [11], equivalently by the endpoint ancient Liouville theorem recorded in Paper I and cited below as [Theorem 9.7](#). Hence $W \equiv 0$. A zero profile has zero pressure-normalized local CKN mass after choosing the admissible pressure gauge, so it cannot be ε_* -concentrating. $\square$

Remark 5.14 (Closure after terminal local concentration analysis). The preceding argument reduces the stationary-hull class with retained local concentration from a time-translation hull question to the question whether a stationary centered profile in a nontrivial stationary family can occur as a Seregin limit. [Theorem 5.10](#) is supplied by the terminal local concentration analysis in the next section.

Remark 5.15 (Recommended statement for the residual decomposition). For the residual decomposition, the third exclusion line should be stated as follows:

$$\begin{array}{c} \text{stationary-hull residual with retained concentration} \\ \xrightarrow{\text{stationary-family Seregin limit exclusion}} \emptyset. \end{array}$$

The precise class is the *degenerate stationary-hull residual class with retained local concentration*.

6. EXCLUSION OF DEGENERATE STATIONARY-HULL PROFILES WITH RETAINED CONCENTRATION

We close the remaining analytic statement in the degenerate stationary-hull class by the terminal local concentration theorem proved in the main paper. The argument is not specific to the existence of a non-isolated stationary family. Once a stationary-family element occurs as a Type I Seregin limit carrying retained local CKN concentration, it is a stationary terminal concentration profile. Such profiles are excluded by the same local terminal concentration analysis and separated-profile exclusion used for the final residual class. No global coercive estimate for the original solution is used.

Throughout this section, a centered Type I profile satisfies

$$(6.1) \quad \partial_\tau U - \Delta U + \frac{1}{2}y \cdot \nabla U + \frac{1}{2}U + (U \cdot \nabla)U + \nabla \Pi = 0, \quad \nabla \cdot U = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$, with the usual local suitability and pressure-gauge conventions.

6.1. Terminal local concentration theorem. We use the following specialization of the terminal local concentration theorem proved below. It is stated here exactly in the form needed for the stationary-hull closure with retained local concentration.

Theorem 6.1 (Terminal local concentration exclusion theorem). *Assume the realized critical-tail rigidity input [Theorem 8.21](#). Let (U, Π) be a bounded centered Type I Seregin profile satisfying local suitability and carrying a retained local CKN concentration, i.e. for some compact parabolic cylinder Q and some $\varepsilon_* > 0$,*

$$(6.2) \quad \iint_Q (|U|^3 + |\Pi - c_Q(\tau)|^{3/2}) \, dy \, d\tau \geq \varepsilon_*.$$

Assume that (U, Π) is outside the previously closed small-amplitude, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, and rotational relative-equilibrium classes. Then the following hold.

- (i) *Every terminal extraction of (U, Π) has the paired decomposition of [Theorem 8.9](#): an active retained profile locus together with a residual remainder which is locally vanishing, diffuse, or noncompact after neighborhoods of the active locus are removed.*
- (ii) *Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the retained compact CKN concentration [\(6.2\)](#).*
- (iii) *Mixed compact-diffuse and compact-noncompact residual states, and infinite active descendant branches, are impossible in the refined residual class.*
- (iv) *Finite separated families of retained concentration profiles are impossible.*
- (v) *Any remaining single retained concentration profile is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no loss of local compactness along escaping recenterings.*
- (vi) *Every retained terminally indecomposable concentration profile has a backward sequence of finite critical norm: there exist $\tau_k \rightarrow -\infty$ such that*

$$(6.3) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

- (vii) *Retained terminal concentration profiles are either affine/parasitic lower-stratum profiles or have parasitic-free finite-shift mild physical pullbacks. Therefore the Albritton–Barker endpoint Liouville theorem applies to any normalized residual profile satisfying [\(6.3\)](#).*

Proof. This is [Theorem 9.20](#) applied to (U, Π) . The local alternatives in (i) are those of [Theorem 8.9](#). Item (ii) is the sole-carrier exclusion in [Theorem 8.12](#). Item (iii) is [Theorems 8.22](#) and [9.16](#). Item (iv) is [Theorem 9.19](#). Item (v) is [Theorem 9.4](#). Item (vi) is [Theorem 9.5](#), and item (vii) is [Theorems 9.6](#) and [9.7](#). The active-locus, active path-space recurrence, and diffuse-defect compactness statements are proved below from local compactness. The only remaining state-space input named in the theorem statement is realized critical-tail rigidity. No global bound is assumed. $\square$

Remark 6.2 (Local nature of the theorem). The theorem above is a local concentration statement. It uses the local suitable-weak-solution compactness and pressure normalization developed from the Caffarelli–Kohn–Nirenberg theory [\[6\]](#), together with the Paper 0 local concentration input and the Paper I Seregin compactness theorem. It then uses paired terminal profile decomposition, sole-carrier exclusions, critical-tail compactification with realized critical-tail rigidity, active path-space recurrence, and the finite separated-profile theorem. It does not assert or require a global a priori bound for the original physical solution.

For completeness, and to make the dependence on local estimates explicit, we record the elementary part of the argument which converts terminal indecomposability into sequence- L^3 control.

Lemma 6.3 (Indecomposable local profiles have backward sequence- L^3 control). *Let (U, Π) be a bounded smooth ancient solution of (6.1). Suppose that (U, Π) has retained local concentration and is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no escaping recentering sequence along which local compactness, suitability, or pressure control fails. Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$(6.4) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We argue by contradiction. Assume that no sequence satisfying (6.4) exists. Then

$$(6.5) \quad \liminf_{\tau \rightarrow -\infty} \|U(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $\int_{\mathbb{R}^3} |U(y, \tau_n)|^3 dy \rightarrow \infty$. Since U is bounded, we may choose radii $R_n \rightarrow \infty$ so that

$$(6.6) \quad \int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

Let

$$(6.7) \quad m_n := \sup_{|x| > R_n} \int_{B_1(x)} |U(y, \tau_n)|^3 dy.$$

If $\limsup_n m_n \geq \varepsilon_*$, then for a subsequence there exist $|x_n| > R_n$ such that

$$(6.8) \quad \int_{B_1(x_n)} |U(y, \tau_n)|^3 dy \geq \varepsilon_*.$$

After the covariant recentering $\mathcal{T}_{(x_n, \tau_n)}$, either the local compactness and pressure control required for a terminal profile fail, or a subsequence converges in the terminal topology to a retained concentrating tail limit. The time-slice lower bound gives parabolic CKN activity by Theorem 9.2. Both alternatives contradict terminal indecomposability.

It remains to consider the case $m_n < \varepsilon_*$ for all sufficiently large n . Then (6.6) says that critical mass persists outside expanding balls, while (6.7) says that every unit-scale exterior recentering remains below the retained-concentration threshold. This is exactly diffuse exterior concentration, again contradicting terminal indecomposability. Thus (6.5) is impossible, and (6.4) follows. $\square$

6.2. Stationary concentration profiles are impossible.

Theorem 6.4 (No stationary residual concentration profile occurs as a Seregin limit). *Assume the realized critical-tail rigidity input Theorem 8.21. Let (W, Π) be a stationary solution of the centered equation,*

$$(6.9) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Assume that:

- (i) (W, Π) is obtained as a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (ii) (W, Π) has retained local concentration in the sense of (6.2);
- (iii) (W, Π) is outside all previously closed lower classes listed in Theorem 6.1;

Then no such (W, Π) exists.

Proof. Suppose that such a stationary profile (W, Π) exists. Since it is a centered Type I Seregin limit and has retained local concentration, it is an admissible input for the terminal local concentration theorem. Apply [Theorem 6.1](#) to (W, Π) .

Finite separated terminal concentration profiles, infinite active descendant branches, and mixed compact–diffuse or compact–noncompact remainders are impossible by [Theorem 6.1](#). Local vanishing, exterior escape, diffuse exterior concentration, and local compactness failure cannot be the only locations of the retained compact CKN concentration. Hence the retained concentration is represented by a single terminal concentration profile with no surviving tail remainder. By the indecomposability conclusion in [Theorem 6.1](#), this single profile is terminally indecomposable.

Because (W, Π) is stationary in τ , the single terminal concentration profile may be represented by (W, Π) itself after the terminal normalization. Therefore [Theorem 6.3](#) gives a sequence $\tau_k \rightarrow -\infty$ such that

$$(6.10) \quad \sup_k \|W\|_{L^3(\mathbb{R}^3)} = \sup_k \|W(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Thus $W \in L^3(\mathbb{R}^3)$.

Let

$$(6.11) \quad w(x, t) = (-t)^{-1/2} W\left(\frac{x}{\sqrt{-t}}\right), \quad q(x, t) = (-t)^{-1} \Pi\left(\frac{x}{\sqrt{-t}}\right), \quad t < 0.$$

By [Theorem 9.6](#), either (W, Π) is in the affine/parasitic lower stratum, already excluded from the residual class, or every finite terminal shift $w^T(x, s) := w(x, T + s)$, $s < 0$, is a bounded mild ancient Navier–Stokes solution. The critical scaling gives

$$(6.12) \quad \|w(\cdot, t)\|_{L^3(\mathbb{R}^3)} = \|W\|_{L^3(\mathbb{R}^3)} < \infty \quad (t < 0).$$

Fix $T < 0$, choose any sequence $t_k \downarrow -\infty$ with $t_k < T$, and put $s_k = t_k - T$. Applying the Albritton–Barker endpoint Liouville theorem to w^T along $s_k \downarrow -\infty$, [Theorem 9.7](#), we obtain $w^T \equiv 0$. Since $T < 0$ is arbitrary, $w \equiv 0$, and hence $W \equiv 0$. This contradicts retained local concentration, since the pressure-normalized local CKN quantity of the zero profile is zero. Therefore no stationary residual concentration profile can occur as such a Seregin limit. $\square$

6.3. Closure of the degenerate stationary-hull residual class.

Corollary 6.5 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. There is no stationary centered profile (W, Π) satisfying all of the following conditions:*

- (i) $W \in \mathfrak{B}$, W solves the stationary centered equation (6.9), and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) has retained local concentration;
- (iv) (W, Π) is outside the previously closed lower classes;

Proof. The stationary-family condition in (i) is additional structure, but the exclusion in [Theorem 6.4](#) applies to every stationary residual profile carrying retained local concentration. Hence it applies in particular to stationary profiles lying in admissible nontrivial stationary families. $\square$

Theorem 6.6 (Closure of the degenerate stationary-hull concentration class). *Assume the realized critical-tail rigidity input [Theorem 8.21](#). Let $\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$ be the degenerate stationary-hull residual class with retained local concentration. Then*

$$(6.13) \quad \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By the definition of the degenerate stationary-hull residual class with retained local concentration, the centered translation hull of (V, P) contains a stationary profile (W, Π) with retained local concentration which lies in an admissible nontrivial stationary family and is outside the previously closed lower classes. The scale-reselection lemma for centered hull limits shows that (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point.

Thus (W, Π) satisfies the hypotheses of [Theorem 6.5](#). This is impossible. Hence no such V exists, and [\(6.13\)](#) follows. $\square$

Remark 6.7 (How this enters the residual decomposition). The third residual line should be recorded as a theorem closed by the terminal local concentration analysis:

$$\begin{aligned} & \text{degenerate stationary-hull residual with retained concentration} \\ & \implies \text{stationary profile with retained concentration} \implies \emptyset. \end{aligned}$$

The second implication is proved by terminal local concentration analysis: decomposable profiles produce either an excluded separated concentration family, diffuse exterior concentration, or loss of local compactness; indecomposable profiles have backward sequence- L^3 control and are ruled out by the endpoint ancient Liouville theorem.

7. TAIL GEOMETRY OF THE GENERIC TERMINAL RESIDUAL

We now pass to the generic terminal residual class. The preliminary point is that the centered equation is not invariant under raw fixed spatial translations. The correct far-field recenterings are self-similar covariant translations.

Definition 7.1 (Covariant translations). For $a \in \mathbb{R}^3$, define

$$(T_a V)(y, \tau) := V(y + e^{\tau/2} a, \tau), \quad (T_a P)(y, \tau) := P(y + e^{\tau/2} a, \tau).$$

Lemma 7.2 (Covariant translation invariance). *If (V, P) solves [\(1.2\)](#), then $(T_a V, T_a P)$ also solves [\(1.2\)](#).*

Proof. Set $z = y + e^{\tau/2} a$. Then

$$\partial_\tau (T_a V)(y, \tau) = (\partial_\tau V)(z, \tau) + \frac{1}{2} e^{\tau/2} a \cdot \nabla V(z, \tau).$$

Moreover,

$$\frac{1}{2} y \cdot \nabla (T_a V)(y, \tau) = \frac{1}{2} z \cdot \nabla V(z, \tau) - \frac{1}{2} e^{\tau/2} a \cdot \nabla V(z, \tau).$$

The two additional transport terms cancel in $\partial_\tau T_a V + \frac{1}{2} y \cdot \nabla T_a V$. The Laplacian, the zeroth-order term, the convection term, the pressure gradient, and the divergence constraint are translation covariant. Substitution into [\(1.2\)](#) proves the claim. $\square$

Remark 7.3 (Raw translates create a spurious drift). If one sets $V_n(y, \tau) = V(y + x_n, \tau)$ with fixed x_n , then the centered equation acquires the additional transport term

$$-\frac{1}{2} x_n \cdot \nabla V_n.$$

Thus fixed translates do not define a compact invariant far-field hull for the renormalized dynamics.

Definition 7.4 (Spacetime covariant action). For $s \in \mathbb{R}$, define the centered time translation

$$(\Theta_s V)(y, \tau) := V(y, \tau + s), \quad (\Theta_s P)(y, \tau) := P(y, \tau + s).$$

For a tail center $x \in \mathbb{R}^3$ observed at centered time σ , define the covariant tail recentering

$$\mathcal{T}_{(x,\sigma)}V(y,s) := V(y + e^{s/2}x, \sigma + s), \quad \mathcal{T}_{(x,\sigma)}P(y,s) := P(y + e^{s/2}x, \sigma + s).$$

Equivalently,

$$\mathcal{T}_{(x,\sigma)} = \Theta_\sigma T_{e^{-\sigma/2}x}.$$

Thus $\mathcal{T}_{(x,\sigma)}$ is a composition of exact symmetries of the centered equation. Raw parabolic recenterings are used below only for local terminal extraction before one has passed to a centered ancient profile; all escaping tail limits of centered profiles use $\mathcal{T}_{(x,\sigma)}$.

Lemma 7.5 (Semidirect covariance of the centered action). *The centered time translations and spatial covariant translations satisfy*

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}, \quad s \in \mathbb{R}, \quad a \in \mathbb{R}^3.$$

In particular the subgroup $\{T_a : a \in \mathbb{R}^3\}$ is normal in the full spacetime covariant action, and the subspace of observables invariant under all spatial covariant translations is invariant under every Θ_s .

Proof. For a profile V ,

$$(\Theta_s T_a \Theta_{-s} V)(y, \tau) = V(y + e^{(\tau+s)/2}a, \tau) = (T_{e^{s/2}a} V)(y, \tau).$$

The pressure transforms identically. The asserted invariance of the spatially invariant observable subspace is the same identity written on the induced L^2 -representation. $\square$

Theorem 7.6 (Covariant observer calculus). *Let $Z := \mathbb{R}^3 \times \mathbb{R}$ denote the centered observer space, with points $z = (\xi, \sigma)$. For $a \in \mathbb{R}^3$ and $\sigma \in \mathbb{R}$,*

$$(T_a U)(y, \tau) = U(y + e^{\tau/2}a, \tau), \quad (\Theta_\sigma U)(y, \tau) = U(y, \tau + \sigma).$$

Then

$$\Theta_\sigma T_a = T_{e^{\sigma/2}a} \Theta_\sigma, \quad \Theta_\sigma T_a \Theta_{-\sigma} = T_{e^{\sigma/2}a}.$$

For $z = (\xi, \sigma)$, define the covariant observer

$$\mathcal{R}_z U := \Theta_\sigma T_{e^{-\sigma/2}\xi} U.$$

Equivalently,

$$(\mathcal{R}_{(\xi,\sigma)} U)(y, s) = U(y + e^{s/2}\xi, \sigma + s).$$

The covariant tube pulled back from $Q_r(0,0)$ is

$$\mathcal{Q}_r^{\text{cov}}(\xi, \sigma) := \left\{ (Y, \tau) : |\tau - \sigma| < r^2, \quad |Y - e^{(\tau-\sigma)/2}\xi| < r \right\}.$$

The map

$$(y, s) \mapsto (Y, \tau) = (y + e^{s/2}\xi, \sigma + s)$$

is a bijection from $Q_r(0,0)$ onto $\mathcal{Q}_r^{\text{cov}}(\xi, \sigma)$ with spacetime Jacobian one. The same formulas apply to the pressure, modulo the admissible time-dependent gauges.

Proof. The semidirect identities are exactly [Theorem 7.5](#); written without conjugation,

$$(\Theta_\sigma T_a U)(y, \tau) = U(y + e^{(\tau+\sigma)/2}a, \tau + \sigma) = (T_{e^{\sigma/2}a} \Theta_\sigma U)(y, \tau).$$

Taking $a = e^{-\sigma/2}\xi$ gives

$$(\Theta_\sigma T_{e^{-\sigma/2}\xi} U)(y, s) = U(y + e^{(s+\sigma)/2}e^{-\sigma/2}\xi, \sigma + s) = U(y + e^{s/2}\xi, \sigma + s).$$

If $Y = y + e^{s/2}\xi$ and $\tau = \sigma + s$, then

$$y = Y - e^{(\tau-\sigma)/2}\xi, \quad s = \tau - \sigma,$$

so $|s| < r^2$ and $|y| < r$ are equivalent to the defining inequalities for $\mathcal{Q}_r^{\text{cov}}(\xi, \sigma)$. The transformation is a time-dependent spatial translation, hence has Jacobian one. $\square$

Definition 7.7 (Terminal state space and local topology). For $M < \infty$, let $\mathfrak{X}_M$ be the class of pairs (U, Π) , modulo addition of arbitrary functions of time to Π , such that:

- (i) U is a smooth bounded ancient solution of (1.2) on $\mathbb{R}^3 \times \mathbb{R}$, with $\|U\|_{L^\infty} \leq M$;
- (ii) (U, Π) is locally suitable on every compact terminal cylinder;
- (iii) on every compact cylinder the pressure has an admissible $L^{3/2}$ -representative after subtracting a time-dependent gauge.

The topology on $\mathfrak{X}_M$ is

$$U_n \rightarrow U \quad \text{in } C_{\text{loc}}^\infty, \quad \Pi_n - c_n(\tau) \rightarrow \Pi \quad \text{in } L_{\text{loc}}^{3/2}$$

for suitable time-dependent gauges c_n . All profile limits, tail descendants, active-tree hulls, and recurrent-core limits below are taken in this topology.

Definition 7.8 (Affine-normalized observer oscillation). Let Q be a bounded covariant terminal cylinder. For a velocity field U set

$$\text{osc}_3(U; Q) := \inf_{c \in \mathbb{R}^3} \iint_Q |U - c|^3 dy d\tau.$$

A covariant observer $z \in Z$ is η -oscillation-active for U if

$$\text{osc}_3(U; \mathcal{Q}_1^{\text{cov}}(z)) \geq \eta.$$

On each compact observer window we fix a uniformly locally finite family of observer centers $\{z_k\} \subset Z$, subordinate to a bounded-overlap covariant unit-cylinder cover. The corresponding local oscillation measure on observer-center sets $E \subset Z$ is

$$\mu_U^{\text{osc}}(E) := \sum_{z_k \in E} \text{osc}_3(U; \mathcal{Q}_1^{\text{cov}}(z_k)).$$

For a sequence U_n we write μ_n^{osc} for this measure. All uses of μ_n^{osc} are on compact observer-center windows or on specified escaping windows after restriction, so the bounded overlap gives uniform local finiteness. Replacing the cover by another bounded-overlap covariant unit cover changes only harmless constants and never changes the alternatives “positive threshold,” “vanishing threshold,” or “diffuse defect.”

Lemma 7.9 (Local compactness and continuity of the state actions). *Let $(U_n, \Pi_n) \in \mathfrak{X}_M$ be a sequence with the local suitability and pressure-gauge bounds supplied by Paper I on every fixed compact cylinder. Then every subsequence has a further subsequence converging in the topology of $\mathfrak{X}_M$. Moreover the actions T_a , Θ_s , the covariant tail recenterings $\mathcal{T}_{(x, \sigma)}$ are continuous on $\mathfrak{X}_M$. The affine maps satisfy the bound-tracking continuity*

$$S_c : \mathfrak{X}_M \longrightarrow \mathfrak{X}_{M+|c|},$$

again with the pressure interpreted modulo the corresponding transformed time gauges.

Proof. The compactness assertion is the standard local Seregini compactness theorem applied on an exhausting sequence of compact cylinders, followed by a diagonal argument. The velocity convergence is upgraded to C_{loc}^∞ for bounded smooth centered profiles by interior parabolic estimates for (1.2); the pressure representatives converge weakly in $L_{\text{loc}}^{3/2}$ after the local gauges. The formulas defining T_a , Θ_s , $\mathcal{T}_{(x, \sigma)}$, and S_c are smooth changes of variables and affine pressure changes on each fixed compact cylinder. The first three preserve the L^∞ bound M . The affine map S_c subtracts the constant vector c , so its velocity bound is at most $M + |c|$. All four maps

preserve the local topology and transform admissible gauges into admissible gauges. No global spatial norm or global pressure representative is used. $\square$

Definition 7.10 (Affine homogeneous symmetry). For $c \in \mathbb{R}^3$, define

$$(S_c V)(y, \tau) := V(y - 2c, \tau) - c, \quad (S_c P)(y, \tau) := P(y - 2c, \tau) + \frac{1}{2}c \cdot y.$$

Lemma 7.11 (Affine renormalized Galilean symmetry). *If (V, P) solves (1.2), then $(S_c V, S_c P)$ also solves (1.2).*

Proof. Set $U(y, \tau) = V(y - 2c, \tau) - c$, $\Pi(y, \tau) = P(y - 2c, \tau) + \frac{1}{2}c \cdot y$, and $z = y - 2c$. Then

$$\partial_\tau U = (\partial_\tau V)(z, \tau), \quad \Delta U = (\Delta V)(z, \tau), \quad \nabla U = (\nabla V)(z, \tau).$$

The nonlinear term is

$$(U \cdot \nabla)U = (V - c) \cdot \nabla V = (V \cdot \nabla)V - c \cdot \nabla V,$$

where all quantities on the right are evaluated at (z, τ) . The drift term satisfies

$$\frac{1}{2}y \cdot \nabla U = \frac{1}{2}(z + 2c) \cdot \nabla V = \frac{1}{2}z \cdot \nabla V + c \cdot \nabla V.$$

The two $c \cdot \nabla V$ terms cancel. Finally,

$$\frac{1}{2}U + \nabla \Pi = \frac{1}{2}(V - c) + \nabla P + \frac{1}{2}c = \frac{1}{2}V + \nabla P.$$

Substitution into (1.2) at the point (z, τ) gives the equation for (U, Π) . The divergence condition is unchanged by translation and subtracting a constant vector. $\square$

Definition 7.12 (Affine-constant lower stratum and residual normalization). The affine-constant stratum consists of centered profiles which become zero after one affine homogeneous symmetry S_c . In the ordered residual decomposition this stratum is part of the previously closed lower classes, not part of $\mathcal{C}_{\text{gen}}$. Equivalently, every generic residual profile is considered after the canonical affine normalization which removes its spatially homogeneous mild mode. Retained activity in the residual class is therefore always measured after this normalization.

Remark 7.13 (How affine constants are used). When a compact recurrent tail core is shown below to consist only of affine constants, the contradiction is not that an unnormalized nonzero constant has zero local L^3 -mass. The contradiction is that affine constants have already been assigned to the closed lower stratum in [Theorem 7.12](#); after the canonical affine normalization they are the zero profile and cannot represent a generic residual state carrying retained activity.

Theorem 7.14 (Closed non-affine activity functional). *Let $Q \Subset \mathbb{R}^3 \times \mathbb{R}$ be a bounded connected parabolic cylinder. For $U \in \mathfrak{X}_M$, define*

$$d_Q(U) := \inf_{c \in \mathbb{R}^3} \left(\iint_Q |U - c|^3 dy d\tau \right)^{1/3}.$$

Then

$$|d_Q(U) - d_Q(V)| \leq \|U - V\|_{L^3(Q)}.$$

In particular d_Q is continuous under the local topology of $\mathfrak{X}_M$. Moreover,

$$d_Q(U) = 0 \iff U \text{ is spatially and temporally constant on } Q.$$

Fix once and for all a nested exhaustion

$$Q_1 \Subset Q_2 \Subset \dots, \quad \bigcup_{j=1}^{\infty} Q_j = \mathbb{R}^3 \times \mathbb{R},$$

by bounded connected parabolic cylinders. Define

$$\Phi(U) := \sum_{j=1}^{\infty} 2^{-j} \min\{1, d_{Q_j}(U)\}.$$

Then Φ is continuous on $\mathfrak{X}_M$, and

$$\Phi(U) = 0 \iff U \text{ is globally constant.}$$

Consequently, for every $\eta > 0$, the non-affine activity set

$$\mathfrak{A}_\eta := \{(U, \Pi) \in \mathfrak{X}_M : \Phi(U) \geq \eta\}$$

is closed in the local state-space topology.

Proof. For every $c \in \mathbb{R}^3$, the triangle inequality in $L^3(Q)$ gives

$$\|U - c\|_{L^3(Q)} \leq \|U - V\|_{L^3(Q)} + \|V - c\|_{L^3(Q)}.$$

Taking the infimum over c gives

$$d_Q(U) \leq \|U - V\|_{L^3(Q)} + d_Q(V).$$

Interchanging U and V proves the Lipschitz estimate. Local smooth convergence implies local L^3 -convergence, so d_Q is continuous in the topology of $\mathfrak{X}_M$.

If $d_Q(U) = 0$, choose $c_k \in \mathbb{R}^3$ such that $\|U - c_k\|_{L^3(Q)} \rightarrow 0$. The sequence c_k is bounded; otherwise

$$\|U - c_k\|_{L^3(Q)} \geq |c_k| |Q|^{1/3} - \|U\|_{L^3(Q)} \rightarrow \infty,$$

which is impossible. Passing to a subsequence, $c_k \rightarrow c$, and hence $U = c$ almost everywhere on Q . Since U is smooth, $U \equiv c$ pointwise on Q . The converse is immediate.

Each summand in the definition of Φ is continuous and bounded by 2^{-j} . The Weierstrass test gives uniform convergence of the defining series, hence continuity of Φ . If $\Phi(U) = 0$, then $d_{Q_j}(U) = 0$ for every j , so U is constant on every Q_j . The cylinders are nested and overlap, so the constants agree and U is globally constant. The converse is again immediate. Closedness of $\mathfrak{A}_\eta$ follows from continuity of Φ . $\square$

Theorem 7.15 (Affine normalization dichotomy). *Let S_c be the affine homogeneous symmetry of Theorem 7.10, and suppose the ordered lower strata include all constant centered profiles as in Theorem 7.12. For every $(U, \Pi) \in \mathfrak{X}_M$ and every $c \in \mathbb{R}^3$, exactly one of the following alternatives holds:*

- (i) $\Phi(S_c U) > 0$, so the affine-normalized representative has positive non-affine retained activity;
- (ii) $\Phi(S_c U) = 0$, so $S_c U$ is globally constant and (U, Π) belongs to the affine/parasitic lower stratum.

Moreover, if $K \subset \mathfrak{X}_M$ is compact and

$$K \cap \{(U, \Pi) : \Phi(U) = 0\} = \emptyset,$$

then there exists $\eta_K > 0$ such that

$$\Phi(U) \geq \eta_K \quad \text{for every } (U, \Pi) \in K.$$

Proof. By Theorem 7.14, $\Phi(S_c U) = 0$ if and only if $S_c U$ is globally constant. That is exactly the affine/parasitic lower stratum recorded in Theorem 7.12. If this does not happen, $\Phi(S_c U) > 0$. The two alternatives are mutually exclusive and exhaust all cases. For the compact statement, Φ is continuous and strictly positive on K , so it attains a positive minimum

$$\eta_K := \min_{(U, \Pi) \in K} \Phi(U) > 0.$$

□

Definition 7.16 (Asymptotic hull at infinity). Let V be a bounded smooth ancient solution of (1.2). Define

$$\mathcal{A}_\infty(V) := \{W : \exists a_n \in \mathbb{R}^3, |a_n| \rightarrow \infty, T_{a_n} V \rightarrow W \text{ in } C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})\}.$$

Theorem 7.17 (Compactness of the asymptotic hull). *Let V be a bounded smooth ancient solution of (1.2). Then $\mathcal{A}_\infty(V)$ is nonempty and compact in $C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})$. Every $W \in \mathcal{A}_\infty(V)$ is a bounded smooth ancient solution of (1.2) with $\|W\|_{L^\infty} \leq \|V\|_{L^\infty}$.*

Proof. For any sequence a_n with $|a_n| \rightarrow \infty$, the translated solutions $T_{a_n} V$ solve the same equation and have the same L^∞ bound by Theorem 7.2. On every compact cylinder, the coefficients of (1.2) are fixed and smooth. Interior parabolic estimates for the Stokes system with bounded smooth nonlinear data, combined with the pressure equation $-\Delta P = \partial_i \partial_j (V_i V_j)$ after local gauges, give uniform C^k -bounds on smaller compact cylinders for every k . A diagonal Arzela–Ascoli argument gives a subsequential limit in C_{loc}^∞ . Passing to the limit in the equation gives another bounded ancient solution. The same compactness argument applied to an arbitrary sequence of elements of $\mathcal{A}_\infty(V)$ proves compactness. □

Definition 7.18 (Tail alternatives). Let $\mathcal{C}$ denote the set of constant vector fields. Define

$$\mathcal{C}_{\text{gentail-const}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \subset \mathcal{C}\},$$

and

$$\mathcal{C}_{\text{gentail-prof}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \text{ contains a nonconstant profile}\}.$$

Proposition 7.19 (Tail dichotomy and tail-zero normalization). *Every $V \in \mathcal{C}_{\text{gen}}$ belongs to $\mathcal{C}_{\text{gentail-const}} \cup \mathcal{C}_{\text{gentail-prof}}$. If $\mathcal{A}_\infty(V) = \{C_\infty\}$ is a singleton constant hull, then*

$$U := S_{C_\infty} V$$

solves (1.2) and satisfies

$$\mathcal{A}_\infty(U) = \{0\}.$$

Proof. The dichotomy is the definition. Since T_a and S_c commute,

$$T_a(S_{C_\infty} V) = S_{C_\infty}(T_a V).$$

Thus every far-field limit of $S_{C_\infty} V$ is

$$S_{C_\infty}(C_\infty) = 0.$$

The equation is preserved by Theorem 7.11. □

Definition 7.20 (Tail-minimal recurrent core). A nonempty compact set $\mathcal{M} \subset \mathcal{A}_\infty(V)$ is a tail-minimal recurrent core if it is invariant under all covariant translations T_a , invariant under all time translations

$$(\Theta_s W)(y, \tau) := W(y, \tau + s),$$

and contains no proper nonempty compact subset with the same two invariance properties.

Lemma 7.21 (Pressure gradients in invariant local averages). *Let $\mathcal{M}$ be a compact set of smooth bounded solutions of (1.2), compact in the locally smooth topology and invariant under the covariant translations T_a and the time translations Θ_s . Let μ be a Borel probability measure on $\mathcal{M}$ invariant under both actions. For each $W \in \mathcal{M}$, let P_W be any smooth local pressure near $(0, 0)$ satisfying (1.2). Put*

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0), \quad b_{ij}(W) := u_i(W)u_j(W).$$

Let Π_0 denote the orthogonal projection in $L^2(\mathcal{M}, \mu)$ onto the closed subspace of functions invariant under all spatial covariant translations T_a , and set

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Then the pressure contribution in the recurrent-core energy average is gauge-independent and satisfies

$$(7.1) \quad \int_{\mathcal{M}} u_i(W) p_i(W) d\mu(W) = \int_{\mathcal{M}} u_i^0(W) p_i^0(W) d\mu(W).$$

Equivalently, the nonzero spatial-frequency part of the pressure gradient does no work in invariant average:

$$(7.2) \quad \int_{\mathcal{M}} u_i(W) (p_i(W) - p_i^0(W)) d\mu(W) = 0.$$

Proof. The additive pressure gauge is irrelevant because only ∇P_W enters. The pressure gradient is fixed by the velocity through the equation

$$\nabla P_W = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - \partial_\tau W - (W \cdot \nabla)W.$$

Thus $W \mapsto p_i(W)$ is a continuous bounded local observable on $\mathcal{M}$. Paper II, [3, Lemma *Local pressure reconstruction*], supplies the local pressure representatives; the present argument uses only their gradients, so the additive gauges never enter.

Let U_a be the unitary representation of spatial covariant translations on $L^2(\mathcal{M}, \mu)$,

$$(U_a F)(W) := F(T_a W).$$

Let D_1, D_2, D_3 be its skew-adjoint infinitesimal generators. For a smooth local observable F , $D_k F$ is obtained by differentiating $F(T_{he_k} W)$ at $h = 0$. In particular, at $\tau = 0$,

$$D_k u_i(W) = \partial_k W_i(0, 0).$$

Hence

$$(7.3) \quad D_i u_i = 0 \quad \text{in } L^2(\mathcal{M}, \mu),$$

because each W is divergence-free.

Taking the spatial divergence of (1.2) gives the local pressure Poisson equation

$$(7.4) \quad -\Delta_y P_W = \partial_i \partial_j (W_i W_j).$$

Indeed, the divergence of $\partial_\tau W$, ΔW , and W is zero; the divergence of $y \cdot \nabla W$ is $\partial_i (y_j \partial_j W_i) = \partial_j W_j + y_j \partial_j \partial_i W_i = 0$; and $(W \cdot \nabla)W_i = \partial_j (W_j W_i)$. Applying the infinitesimal translation generators at $(0, 0)$ gives

$$(7.5) \quad D_i p_i = -D_i D_j b_{ij}.$$

The pressure gradient is curl-free, hence

$$(7.6) \quad D_k p_i = D_i p_k.$$

Let

$$A := -\sum_{k=1}^3 D_k^2.$$

By the spectral theorem for the commuting unitary group U_a , A is self-adjoint and nonnegative, and Π_0 is exactly the spectral projection of A at the eigenvalue 0. Define the nonzero spatial part

$$q_i := p_i - p_i^0 = (I - \Pi_0)p_i.$$

Equations (7.5) and (7.6) imply, on the nonzero spectral subspace,

$$q_k = D_k A^{-1} (I - \Pi_0) D_i D_j b_{ij}$$

in the standard regularized sense. Explicitly, with

$$q_{k,\varepsilon} := D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij}, \quad \varepsilon > 0,$$

the joint spectral resolution of D_1, D_2, D_3 gives

$$q_{k,\varepsilon} \longrightarrow q_k \quad \text{in } L^2(\mathcal{M}, \mu) \quad \text{as } \varepsilon \downarrow 0.$$

The convergence is just the multiplier convergence

$$\frac{\xi_k \xi_i \xi_j}{\varepsilon + |\xi|^2} \longrightarrow \frac{\xi_k \xi_i \xi_j}{|\xi|^2} \quad (\xi \neq 0),$$

together with $(I - \Pi_0)$, which removes the point $\xi = 0$. The curl-free relation (7.6) says that the nonzero spatial spectral part of p is parallel to ξ , and (7.5) fixes its longitudinal coefficient; there is therefore no additional harmonic component.

For each $\varepsilon > 0$, skew-adjointness of D_k and (7.3) give

$$\begin{aligned} \int_{\mathcal{M}} u_k q_{k,\varepsilon} d\mu &= \int_{\mathcal{M}} u_k D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu \\ &= - \int_{\mathcal{M}} (D_k u_k) (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu = 0. \end{aligned}$$

Passing to the limit in L^2 yields

$$\int_{\mathcal{M}} u_k q_k d\mu = 0,$$

which is (7.2). Since p_i^0 lies in the range of Π_0 and $u_i - u_i^0$ is orthogonal to that range,

$$\int_{\mathcal{M}} (u_i - u_i^0) p_i^0 d\mu = 0.$$

Combining this with (7.2) proves (7.1). $\square$

Theorem 7.22 (Rigidity of recurrent profile tail cores). *Let $\mathcal{M}$ be a compact tail-minimal recurrent core for (1.2) arising from the Paper I normalized Seregin theorem, with the affine zero mode interpreted by the affine symmetry Theorem 7.11. Equivalently, after applying one fixed affine normalization S_c , the spatially homogeneous part of the core satisfies the bounded mild centered formulation of Paper II [3, Definition Bounded mild ancient solution]. Then every $W \in \mathcal{M}$ is spatially and temporally constant. More precisely, there exists $m \in \mathbb{R}^3$ such that*

$$W(y, \tau) \equiv m \quad \text{for every } W \in \mathcal{M}.$$

After applying the affine symmetry S_m , the core is reduced to the zero solution.

Proof. The group generated by covariant translations and time translations is amenable on the compact invariant set $\mathcal{M}$. Choose an invariant Borel probability measure μ with full support on a minimal component, and write

$$\langle F \rangle := \int_{\mathcal{M}} F(W) d\mu(W)$$

for local observables. Let P_W be a local pressure associated to W . As in Theorem 7.21, set

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0),$$

and let Π_0 be the orthogonal projection onto observables invariant under all spatial covariant translations. Write

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Invariance gives

$$\langle \partial_\tau F(W)(0,0) \rangle = 0, \quad \langle \partial_j F(W)(0,0) \rangle = 0$$

for every smooth local observable F . These identities follow first for difference quotients generated by the group action and then by smoothness.

Evaluate (1.2) at $(0,0)$. The drift term vanishes at $y = 0$, and we get

$$\partial_\tau W(0,0) - \Delta W(0,0) + \frac{1}{2}W(0,0) + (W(0,0) \cdot \nabla)W(0,0) + \nabla P_W(0,0) = 0.$$

Project this identity onto the spatially invariant subspace. The spatial derivative terms vanish under Π_0 : the Laplacian is $D_j D_j u_i$, and the nonlinear term is $D_j(u_j u_i)$ because $D_j u_j = 0$. Time translations do not literally commute with the covariant spatial translations; by Theorem 7.5 they normalize them:

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}.$$

Hence the subspace of observables invariant under all T_a is invariant under Θ_s , and the orthogonal projection Π_0 commutes with the time-translation representation. Therefore $\Pi_0 \partial_\tau u_i = \partial_\tau u_i^0$. Thus

$$(7.7) \quad \partial_\tau u_i^0 + \frac{1}{2}u_i^0 + p_i^0 = 0 \quad \text{in } L^2(\mathcal{M}, \mu).$$

Taking the inner product of (7.7) with u_i^0 , summing in i , and using invariance under time translations,

$$\langle u_i^0 \partial_\tau u_i^0 \rangle = \frac{1}{2} \langle \partial_\tau |u^0|^2 \rangle = 0,$$

gives

$$(7.8) \quad \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Dot the equation with W , evaluate at $(0,0)$, and average. The time term is

$$\langle W \cdot \partial_\tau W \rangle = \frac{1}{2} \langle \partial_\tau |W|^2 \rangle = 0.$$

The diffusion term gives

$$-W \cdot \Delta W = |\nabla W|^2 - \operatorname{div}(W \cdot \nabla W),$$

and therefore

$$-\langle W \cdot \Delta W \rangle = \langle |\nabla W|^2 \rangle.$$

The nonlinear term is a divergence:

$$W \cdot (W \cdot \nabla)W = \frac{1}{2}W \cdot \nabla |W|^2 = \frac{1}{2} \operatorname{div}(|W|^2 W),$$

so its average is zero. Decompose

$$p_i = p_i^0 + (p_i - p_i^0).$$

The nonzero spatial part of the pressure gradient does no work in invariant average by Theorem 7.21, and the remaining spatial zero mode is evaluated by (7.8). Hence

$$\langle u_i p_i \rangle = \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Thus

$$(7.9) \quad \langle |\nabla W|^2 \rangle + \frac{1}{2} \langle |W|^2 \rangle - \frac{1}{2} \langle |u^0|^2 \rangle = 0.$$

Both terms are nonnegative. The second nonnegative quantity is

$$\langle |u|^2 \rangle - \langle |u^0|^2 \rangle = \langle |u - u^0|^2 \rangle,$$

because Π_0 is an orthogonal projection in $L^2(\mathcal{M}, \mu)$. Therefore

$$\langle |\nabla W|^2 \rangle = 0, \quad \langle |u - u^0|^2 \rangle = 0.$$

Since μ has full support and the observables are continuous in the local smooth topology, $\nabla W(0, 0) = 0$ for every $W \in \mathcal{M}$. Invariance moves $(0, 0)$ to any (y, τ) , so $\nabla W(y, \tau) = 0$ everywhere. Thus every element of $\mathcal{M}$ is spatially constant:

$$W(y, \tau) = a_W(\tau).$$

It remains only to identify the allowed spatially homogeneous zero mode. This is not a pressure-gauge assertion. By the Paper I normalization quoted in the statement, there is one fixed affine normalization S_c for which the spatially homogeneous representatives satisfy the bounded mild centered formulation of Paper II. For a spatially constant mild centered solution $B(\tau)$, the nonlinear Duhamel term in Paper II's mild formula vanishes and the Ornstein–Uhlenbeck–Stokes semigroup of Paper II acts on constants by

$$S(\tau - \sigma)B(\sigma) = e^{-(\tau - \sigma)/2}B(\sigma).$$

Thus

$$B(\tau) = e^{-(\tau - \sigma)/2}B(\sigma) \quad (\sigma < \tau).$$

Letting $\sigma \rightarrow -\infty$ and using boundedness of the ancient trajectory gives $B(\tau) = 0$ for every τ . Undoing the fixed affine normalization gives a single vector $m = c$ and

$$W(y, \tau) \equiv m \quad (W \in \mathcal{M}).$$

The affine symmetry S_m then maps the core to the zero solution. $\square$

8. TERMINAL CONCENTRATION PROFILES AND SEPARATED CENTERS

The terminal part of the proof is local in parabolic cylinders. It does not require a global L^3 bound, a global CKN budget, a finite critical-mass budget, or a global nonlinear profile decomposition for the original residual profile. Everything is formulated on fixed compact terminal cylinders after applying the recentering sequence under consideration.

Definition 8.1 (Parabolic recentering sequence). Let (V_n, P_n) be a terminal extraction sequence on expanding domains $D_n \uparrow \mathbb{R}^3 \times \mathbb{R}$, with local suitability and pressure-gauge control. A parabolic recentering sequence is a sequence

$$\mathbf{c} = (z_n)_{n \geq 1}, \quad z_n = (y_n, \tau_n) \in D_n.$$

Two recentering sequences are equivalent if

$$\sup_n \left(|y_n - y'_n| + |\tau_n - \tau'_n|^{1/2} \right) < \infty,$$

and asymptotically separated if the same parabolic distance tends to infinity. The recentered fields associated to $\mathbf{c}$ are denoted

$$V_n^{\mathbf{c}}(y, s) := V_n(y + y_n, s + \tau_n), \quad P_n^{\mathbf{c}}(y, s) := P_n(y + y_n, s + \tau_n) - a_n^{\mathbf{c}}(s),$$

where $a_n^{\mathbf{c}}$ is an admissible time-dependent local pressure gauge. All convergence assertions for a recentering sequence are made on fixed compact sets in the variables (y, s) .

Definition 8.2 (Combined raw-covariant descendant recentering). Let $g_n = (y_n, \tau_n)$ be a raw parabolic recentering sequence for a terminal extraction (V_n, P_n) , and let $\zeta_n = (\xi_n, \sigma_n)$ be a covariant tail recentering sequence in the centered limit generated by g_n . The combined raw-covariant descendant recentering is denoted

$$d_n = (g_n; \xi_n, \sigma_n)$$

and acts on the original terminal sequence by

$$V_n^d(y, s) := V_n(y + y_n + e^{s/2}\xi_n, s + \tau_n + \sigma_n),$$

with the pressure pulled back by the same formula and then adjusted by an admissible time-dependent gauge. Its relative distance from g_n is

$$|\xi_n| + |\sigma_n|^{1/2}.$$

Thus d_n is separated from g_n exactly when this relative distance tends to infinity. This definition is used only to lift descendants of a centered profile back to the original terminal extraction; ordinary first-generation terminal recenterings remain the raw parabolic recenterings of [Theorem 8.1](#).

Definition 8.3 (Local CKN measure and concentrating recentering sequences). After choosing local pressure gauges $a_n(\tau)$, define

$$d\mu_n := \left(|V_n|^3 + |P_n - a_n(\tau)|^{3/2} \right) dy d\tau, \quad d\nu_n := |\nabla V_n|^2 dy d\tau.$$

The local mass of a recentering sequence $\mathbf{c} = (z_n)$ is

$$m(\mathbf{c}) := \limsup_{n \rightarrow \infty} \mu_n(Q_1(z_n)).$$

For a combined raw-covariant descendant recentering $d_n = (g_n; \xi_n, \sigma_n)$, the same notation means the mass of the pulled back CKN measure in the combined coordinates:

$$m(\mathbf{d}) := \limsup_{n \rightarrow \infty} \mu_n^{\mathbf{d}}(Q_1(0, 0)),$$

where $\mu_n^{\mathbf{d}}$ is obtained from μ_n by the coordinate change in [Theorem 8.2](#) and by the admissible pressure gauges used in that recentered frame. Thus raw first-generation recentering sequences and lifted covariant descendants use the same local unit-cylinder criterion, expressed in their own local coordinates. For a fixed small-data threshold $\varepsilon_{\text{sd}} > 0$, the recentering sequence is ε_{sd} -concentrating if $m(\mathbf{c}) \geq \varepsilon_{\text{sd}}$, and locally vanishing at unit scale otherwise. It has loss of local compactness if

$$\limsup_{n \rightarrow \infty} \nu_n(Q_2(z_n)) = \infty$$

or if the local pressure gauges fail to be uniformly controlled in $L^{3/2}(Q_2(z_n))$.

Lemma 8.4 (Gauge compatibility and retained-activity semicontinuity). *Let $(U_n, \Pi_n) \rightarrow (U, \Pi)$ in the topology of $\mathfrak{X}_M$ on a fixed compact cylinder containing $Q_1(0, 0)$. After choosing the pressure gauges used in that convergence, the following hold.*

- (i) *The pressure gauges are compatible with centered time shifts, covariant translations, covariant tail recenterings, affine normalization, and diagonal profile-tree extraction: applying any of these operations only changes the gauge by another admissible function of time.*
- (ii) *The local CKN quantity is lower semicontinuous in the direction needed for lifting active limits:*

$$\mathcal{E}_1(U, \Pi; 0) \leq \liminf_{n \rightarrow \infty} \mathcal{E}_1(U_n, \Pi_n; 0).$$

Consequently, if the limit profile has $\mathcal{E}_1(U, \Pi; 0) \geq \eta$, then all sufficiently far approximating profiles have CKN mass at least any prescribed $\eta' < \eta$.

- (iii) *If retained activity is carried by the velocity part, then it is closed under the same convergence:*

$$\liminf_{n \rightarrow \infty} \iint_{Q_1} |U_n|^3 \geq \eta \quad \implies \quad \iint_{Q_1} |U|^3 \geq \eta.$$

In applications where the lower bound is first obtained for velocity alone and then converted into a CKN lower bound, the descendant threshold is chosen below the resulting positive constant as in [Theorem 2.2](#).

Proof. Time shifts and covariant translations compose the pressure with a smooth local change of variables; affine normalization adds the explicit linear pressure $\frac{1}{2}c \cdot y$; and every local pressure is defined only up to a time-dependent function. Thus admissible gauges remain admissible under all operations listed in (i). For (ii), strong local convergence of U_n gives

$$\iint_{Q_1} |U_n|^3 \rightarrow \iint_{Q_1} |U|^3.$$

The pressure terms are convex in the weak $L^{3/2}$ topology after the fixed gauges, hence

$$\iint_{Q_1} |\Pi - c(\tau)|^{3/2} \leq \liminf_{n \rightarrow \infty} \iint_{Q_1} |\Pi_n - c_n(\tau)|^{3/2}.$$

Adding the two inequalities gives the claimed lower semicontinuity direction for $\mathcal{E}_1$. The velocity-only assertion follows directly from strong $L^3(Q_1)$ convergence. The statement is entirely local on Q_1 . $\square$

Lemma 8.5 (Paper I compactness excludes retained local compactness failure). *Let (V_n, P_n) be a terminal sequence obtained from a Paper I Type I Seregin extraction, or from a later terminal extraction of such a sequence. If a recentering sequence $\mathfrak{c} = (z_n)$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness and its recentered fields converge locally in the Seregin topology to a suitable bounded terminal profile.*

Proof. Fix $R < \infty$. The recentered cylinders $Q_{2R}(z_n)$ are compact terminal cylinders in the variables of the chosen extraction. Paper I gives local suitability, local pressure-gauge compactness, and the inherited Type I bound on every such compact cylinder. Therefore, after choosing the pressure gauges $a_n^\mathfrak{c}$, the sequence $(V_n^\mathfrak{c}, P_n^\mathfrak{c})$ has the standard local bounds needed for the Seregin compactness theorem on $Q_R(0, 0)$: strong local L^3 compactness for the velocity, weak local $L^{3/2}$ compactness for the pressure, and the local energy inequality in the limit. A diagonal argument over $R = 1, 2, \dots$ gives a suitable bounded terminal profile. The two failures used in the definition of loss of local compactness are precisely the negations of these local compactness and pressure-gauge conclusions, so a retained recentering sequence arising from the Paper I Type I extraction theorem cannot remain in the local-compactness-failure alternative. $\square$

Theorem 8.6 (Local active-locus extraction). *Let $Z = \mathbb{R}^3 \times \mathbb{R}$ be the centered covariant observer space, and let $K_m \Subset Z$, $m \geq 1$, be a compact exhaustion. Let μ_n be the local CKN measures of a terminal sequence, with local finite mass on compact observer windows:*

$$\sup_n \mu_n(K_m^+) < \infty \quad (m \geq 1),$$

where K_m^+ denotes the union of the supports of the transported bumps defined below for $z \in K_m$. Fix $\chi \in C_c^\infty(Q_1(0, 0))$, $0 \leq \chi \leq 1$, with $\chi \equiv 1$ on $Q_{1/2}(0, 0)$. For $z \in Z$, put

$$\chi_z(Y, \tau) := \chi(\mathcal{R}_z^{-1}(Y, \tau)), \quad M_n(z) := \int \chi_z d\mu_n.$$

For $\eta > 0$, define the compact-window active locus

$$A_{n,m}^\eta := \{z \in K_m : M_n(z) \geq \eta\}.$$

After passing to a subsequence, for every m and every rational $\eta > 0$, the compact sets $A_{n,m}^\eta$ converge in the Hausdorff–Fell topology to compact sets $A_m^\eta \subset K_m$. Moreover, for every open

neighborhood $O \subset K_m$ of A_m^η ,

$$\limsup_{n \rightarrow \infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta.$$

Thus outside every neighborhood of the active locus A_m^η , no η -active covariant unit tube remains in the compact observer window K_m . The active object is the closed locus A_m^η , not a finite or countable enumeration of tubes.

Proof. For fixed n , the map $z \mapsto \chi_z$ is continuous in C_c on K_m , and all supports are contained in the compact enlargement K_m^+ . Since μ_n is a Radon measure finite on K_m^+ , M_n is continuous on K_m . Therefore $A_{n,m}^\eta$ is compact.

The hyperspace of compact subsets of the compact metric space K_m is compact in the Hausdorff–Fell topology. Hence, for each fixed pair (m, η) , a subsequence exists along which $A_{n,m}^\eta$ converges. A diagonal argument over $m \in \mathbb{N}$ and $\eta \in \mathbb{Q}_+$ gives simultaneous convergence.

Let $O \subset K_m$ be an open neighborhood of A_m^η . If the stated residual bound failed, there would exist $\delta > 0$, a subsequence n_k , and points $z_{n_k} \in K_m \setminus O$ such that

$$M_{n_k}(z_{n_k}) \geq \eta + \delta.$$

After replacing O by a slightly smaller open neighborhood of A_m^η , we may assume $K_m \setminus O$ is compact. Passing to a further subsequence,

$$z_{n_k} \rightarrow z_* \in K_m \setminus O.$$

For all large k ,

$$z_{n_k} \in A_{n_k,m}^{\eta+\delta/2}.$$

Hausdorff–Fell convergence gives $z_* \in A_m^{\eta+\delta/2}$. Since $A_m^{\eta+\delta/2} \subset A_m^\eta \subset O$, this contradicts $z_* \in K_m \setminus O$. The residual estimate follows. All bounds used here are local on K_m^+ ; no global CKN budget or countability of active profiles is assumed. $\square$

Lemma 8.7 (Oscillation-active observers are CKN-active). *Let μ_n be the local CKN measure used in Theorem 8.6, so that μ_n dominates the velocity measure $|U_n|^3 dy d\tau$. If a covariant unit observer z satisfies*

$$\text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) \geq \eta,$$

then the same observer is CKN-active at threshold η :

$$\mu_n(\mathcal{Q}_1^{\text{cov}}(z)) \geq \eta.$$

Consequently, after the CKN active loci of Theorem 8.6 are removed at every positive threshold, no positive affine-normalized oscillation-active unit observer remains outside those loci.

Proof. By definition of the affine-normalized oscillation,

$$\text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) = \inf_{c \in \mathbb{R}^3} \iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n - c|^3 \leq \iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n|^3.$$

The local CKN measure contains the velocity contribution, hence

$$\iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n|^3 \leq \mu_n(\mathcal{Q}_1^{\text{cov}}(z)).$$

Thus oscillation activity implies CKN activity with the same threshold. The last statement follows by applying the local active-locus extraction theorem to all rational positive thresholds and then lowering thresholds by harmless fixed cover and cutoff constants when the smoothed bump convention is used. $\square$

Lemma 8.8 (Separation of recentered cylinders). *Let $\mathbf{c}_i = (z_{i,n})$ and $\mathbf{c}_j = (z_{j,n})$ be asymptotically separated recentering sequences. For every fixed $R < \infty$, the cylinder $Q_R(z_{j,n})$, written in the coordinates of $\mathbf{c}_i$, leaves every compact subset of the recentered variables. Consequently no positive compact velocity L^3 -mass assigned to $\mathbf{c}_j$ is visible on any fixed compact cylinder of the $\mathbf{c}_i$ -frame.*

Proof. In the $\mathbf{c}_i$ -coordinates the center of $Q_R(z_{j,n})$ is

$$(y_{j,n} - y_{i,n}, \tau_{j,n} - \tau_{i,n}).$$

The parabolic distance of this point from the origin tends to infinity by asymptotic separation. Hence, for every fixed compact terminal cylinder K , one has

$$K \cap (Q_R(z_{j,n}) - z_{i,n}) = \emptyset$$

for all sufficiently large n . This is a purely local statement about the relative positions of the two recentering sequences. It does not estimate the solution on all of space and does not use a global norm. $\square$

Theorem 8.9 (Paired terminal active-profile decomposition). *Every terminal concentration sequence admits a paired decomposition*

$$\text{active retained profile locus} + \text{residual remainder}.$$

More precisely, after passing to a subsequence, the compact observer-window active loci A_m^η of Theorem 8.6 exist simultaneously for every m and every rational $\eta > 0$. Every observer sequence that remains in these loci at a fixed positive threshold and has no loss of local compactness generates a bounded retained terminal profile after local pressure gauges. Outside every neighborhood of the active locus in each fixed compact observer window, the residual local CKN mass is below the chosen threshold. The residual remainder is then of exactly one of the following types:

$$\mathcal{R}_{\text{van}}, \quad \mathcal{R}_{\text{diff}}, \quad \mathcal{R}_{\text{noncomp}}.$$

Here $\mathcal{R}_{\text{van}}$ means no critical mass remains on expanding regions, $\mathcal{R}_{\text{diff}}$ means diffuse exterior critical mass remains on expanding regions while every fixed unit cylinder is below threshold, and $\mathcal{R}_{\text{noncomp}}$ means that some escaping residual recentering loses the local Seregin compactness or pressure-gauge control. Thus mixed states such as one active compact profile plus diffuse exterior remainder are part of the decomposition; they are not silently identified with the pure diffuse stratum.

Proof. Apply the local active-locus extraction theorem, Theorem 8.6, on the compact observer exhaustion $\{K_m\}$. It produces closed active loci A_m^η and the residual estimate

$$\limsup_{n \rightarrow \infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta$$

outside every neighborhood O of A_m^η . This is the precise replacement for a maximal finite or countable tube family: active centers are recorded by closed loci, and the residual is defined by restricting away from their neighborhoods.

If an observer sequence stays in a positive-threshold active locus and has no loss of the local compactness specified in Theorem 8.3, then Theorem 8.5 and the standard diagonal Seregin compactness argument give a bounded retained terminal profile. If such an observer sequence loses local dissipation or pressure-gauge control, the corresponding remainder is $\mathcal{R}_{\text{noncomp}}$.

It remains to classify what is outside the active loci. If the residual critical CKN mass tends to zero on every expanding terminal region, the remainder is $\mathcal{R}_{\text{van}}$. If a positive amount of critical CKN mass persists on expanding regions while every fixed compact observer window is below every active threshold after removing neighborhoods of the active loci, the remainder is $\mathcal{R}_{\text{diff}}$. These three residual states are mutually exclusive once the active loci are fixed, and they exhaust

the residual possibilities. No step asserts that a diffuse or exterior remainder cannot coexist with active compact profiles. $\square$

Corollary 8.10 (Persistent local velocity mass forces retained concentration). *If a terminal sequence satisfies*

$$\liminf_{n \rightarrow \infty} \iint_{Q_1(0,0)} |V_n|^3 dy d\tau \geq \eta_0 > 0$$

with $\eta_0 \geq \varepsilon_{\text{sd}}$, then the terminal decomposition contains a local concentration profile unless the distinguished observer sequence has loss of local compactness. In the generic residual after local vanishing and local compactness failure are excluded, the persistent compact velocity mass is therefore carried by the active retained profile locus of [Theorem 8.9](#); the locus may contain one, finitely many, infinitely many, or continuum many active observer classes, and the residual remainder may still be diffuse until the mixed compact–diffuse stratum is excluded.

Proof. The local CKN measure μ_n dominates the velocity contribution. Hence the origin recentering sequence has μ -mass at least η_0 , and is ε_{sd} -concentrating unless it has loss of local compactness. If it has no loss of local compactness, active-locus extraction places its observer class in a positive-threshold active locus A_m^η of [Theorem 8.9](#). $\square$

Lemma 8.11 (Local vanishing and exterior alternatives do not carry compact concentration). *Let a terminal extraction inherit the positive compact velocity L^3 -mass supplied by Paper 0 and Paper I. Then the retained compact velocity mass cannot be carried solely by local vanishing, exterior escape, or diffuse exterior concentration.*

Proof. The assertion is local. By Paper 0, failure of local vanishing at a singular terminal point gives a positive local velocity concentration sequence. Paper I carries this through the Type I Seregin extraction as the compact velocity lower bound (1.5). Cover the fixed compact cylinder K in (1.5) by finitely many unit terminal cylinders. At least one cylinder in the cover carries a positive velocity lower bound, say $\eta_K > 0$. Since μ_n dominates $|V_n|^3 dy d\tau$, choosing $\varepsilon_{\text{sd}} \leq \eta_K$ makes that unit cylinder ε_{sd} -concentrating. Thus the distinguished compact concentration cannot be locally vanishing.

Local vanishing after removal of concentrating recentering sequences means that the residual state has local CKN mass below ε_{sd} on each fixed compact terminal cylinder; it therefore cannot contain the fixed compact velocity lower bound. An exterior escaping concentration is visible only after its own recentering; relative to the distinguished compact terminal cylinder it escapes every bounded window. By [Theorem 8.8](#), it cannot be the sole location of a lower bound that remains in the distinguished compact window. Finally, diffuse exterior concentration is defined by persistence through expanding regions while all fixed unit cylinders are locally vanishing. If such a sequence contained the compact Paper 0/Paper I velocity lower bound, the positive concentration lemma of Paper 0 would select a unit terminal cylinder with mass at least the retained threshold, contradicting local vanishing. Thus none of these alternatives can be the unique location of the retained compact velocity mass. $\square$

Theorem 8.12 (Local exclusion of non-compact and non-concentrating terminal alternatives). *For terminal sequences generated by the Paper I Type I Seregin extraction theorem, loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are not possible sole locations of the retained compact velocity mass.*

Proof. Retained recentering sequences with loss of local compactness are removed by [Theorem 8.5](#). Local vanishing, exterior escape, and diffuse exterior concentration are removed by [Theorem 8.11](#). The proof uses only the local positive concentration theorem of Paper 0, the local Type I Seregin compactness and pressure gauges of Paper I, and fixed compact terminal cylinders. $\square$

Theorem 8.13 (Diffuse-defect compactness). *Let μ_n^{osc} be the residual affine-normalized oscillation measures of Theorem 7.8, after removing neighborhoods of all positive-threshold active loci from Theorem 8.6. By Theorem 8.7, this removal also removes all positive-threshold oscillation-active unit observers on compact windows. Assume there are observer regions $E_n \subset Z$, escaping every compact subset of Z , and a number $m_0 > 0$ such that*

$$\mu_n^{\text{osc}}(E_n) \geq m_0,$$

while no unit observer in E_n carries a positive oscillation threshold:

$$\sup_{z \in E_n} \text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) \longrightarrow 0.$$

Let $\bar{Z}$ be a compact metrizable observer compactification on which the fixed covariant observer maps extend continuously, and define probability measures

$$\lambda_n := \frac{\mu_n^{\text{osc}} \lfloor E_n}{\mu_n^{\text{osc}}(E_n)} \quad \text{on } \bar{Z}.$$

After passing to a subsequence,

$$\lambda_n \rightharpoonup \lambda$$

weakly as probability measures on $\bar{Z}$. The limit satisfies

$$\lambda(\bar{Z}) = 1, \quad \lambda(B) = 0 \quad \text{for every } B \Subset Z.$$

Thus the limit is a nonzero diffuse-defect state supported at the boundary of the observer compactification, not a local active profile.

Proof. The measures λ_n are probability measures on the compact metric space $\bar{Z}$, so weak compactness gives a convergent subsequence. Testing against the constant function 1 preserves total mass and gives $\lambda(\bar{Z}) = 1$.

Let $B \Subset Z$. Since E_n escapes every compact subset of Z , one has $B \cap E_n = \emptyset$ for all sufficiently large n . Hence $\lambda_n(B) = 0$ eventually, and the portmanteau theorem gives $\lambda(B) = 0$. The additional unit-scale diffuseness assumption gives the stronger moving-window estimate

$$\sup_{z \in E_n} \lambda_n(B_Z(z, 1)) \leq \frac{C}{m_0} \sup_{\zeta \in E_n} \text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(\zeta)) \longrightarrow 0.$$

Here $B_Z(z, 1)$ is any fixed unit ball for a proper metric on Z , and C is the bounded-overlap constant of the observer cover. The restriction $\lambda_n = \mu_n^{\text{osc}} \lfloor E_n / \mu_n^{\text{osc}}(E_n)$ is why only cover centers lying in E_n enter the estimate. Therefore no positive unit-scale local oscillation profile is hidden inside the escaping windows. The proof is local on the chosen observer windows and uses no global critical-mass or energy budget. $\square$

Definition 8.14 (Diffuse-defect state space). Let $\mathfrak{D}_M$ be the compact closure, in the weak topology of probability measures on $\bar{Z}$ together with the local $\mathfrak{X}_M$ -topology for the associated velocity hulls, of all normalized diffuse-defect states produced by Theorem 8.13. The diffuse-defect mass functional is normalized to be identically one on $\mathfrak{D}_M$. This compactness is the product of weak compactness of probability measures on the compact observer space $\bar{Z}$ and the local Seregin compactness of bounded velocity hulls in $\mathfrak{X}_M$, after the admissible pressure gauges. The covariant observer group

$$G := \mathbb{R}^3 \rtimes \mathbb{R}$$

acts on $\mathfrak{D}_M$ by the observer maps generated by T_a and Θ_s .

Theorem 8.15 (Recurrent diffuse-defect measure). *Assume the covariant observer action of G is continuous on $\mathfrak{D}_M$. Every nonzero diffuse-defect state $\lambda \in \mathfrak{D}_M$ generates a G -invariant Borel probability measure $\mathbb{P}$ on $\mathfrak{D}_M$. The normalized diffuse-defect mass remains one under this averaging.*

Proof. The group $G = \mathbb{R}^3 \rtimes \mathbb{R}$, with the semidirect law from [Theorem 7.5](#), is solvable and hence amenable. Let (F_α) be a left Følner net for G , with Haar measure $|F_\alpha|$, and define probability measures on $\mathfrak{D}_M$ by

$$\mathbb{P}_\alpha := \frac{1}{|F_\alpha|} \int_{F_\alpha} \delta_{g\lambda} dg.$$

Compactness of $\mathfrak{D}_M$ implies weak-* compactness of its probability measures, so a subnet converges weakly to a probability measure $\mathbb{P}$. For every continuous observable F on $\mathfrak{D}_M$ and every fixed $h \in G$,

$$\int F(h\omega) d\mathbb{P}_\alpha(\omega) - \int F(\omega) d\mathbb{P}_\alpha(\omega) = \frac{1}{|F_\alpha|} \int_{F_\alpha} (F(hg\lambda) - F(g\lambda)) dg.$$

The Følner property makes the right-hand side tend to zero. Passing to the weak limit gives G -invariance of $\mathbb{P}$. Since the diffuse-defect mass is identically one on $\mathfrak{D}_M$, its average against $\mathbb{P}$ is also one. $\square$

Theorem 8.16 (Diffuse-defect trichotomy). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a normalized terminal residual sequence with $\|U_n\|_{L^\infty} \leq M$. Suppose that, after removing the active loci at every positive affine-normalized threshold, there are escaping observer windows E_n such that*

$$\mu_n^{\text{osc}}(E_n) \geq m_0 > 0.$$

Then, after passing to a subsequence, the residual diffuse branch falls into one of the following ordered alternatives:

D1. Active regeneration. *There exist covariant observers z_n and a threshold $\eta > 0$ such that*

$$\text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z_n)) \geq \eta.$$

This produces a retained active descendant.

D2. Affine/parasitic collapse. *Every velocity tangent associated with the diffuse-defect hull is locally constant after the admissible affine normalization; the branch belongs to the affine/parasitic lower stratum of [Theorem 7.12](#).*

D3. Critical diffuse tail. *There is a nonzero recurrent diffuse-defect state supported on $\overline{Z} \setminus Z$, carrying normalized diffuse oscillation mass one, annihilating every fixed unit active observable, and not contained in the affine/parasitic lower stratum.*

Equivalently, residual diffuse defects are contained in the union of the active regeneration branch, the affine/parasitic branch, and the critical diffuse-tail class $\mathcal{C}_{\text{critdiff}}$.

Proof. If D1 occurs there is nothing to prove. Assume D1 does not occur. Then, for every fixed $\eta > 0$, all sufficiently large n satisfy

$$\sup_{z \in E_n} \text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) < \eta.$$

The hypotheses of [Theorem 8.13](#) apply to $\mu_n^{\text{osc}}|_{E_n}$, and hence produce a diffuse-defect state λ with total mass one and zero mass on each compact subset of Z . By [Theorem 8.15](#), the orbit closure of λ carries a G -invariant probability measure $\mathbb{P}$.

If every local velocity tangent in the support of $\mathbb{P}$, after the canonical affine normalization, is locally constant, then [Theorem 7.12](#) places the branch in the affine lower stratum. This is D2. If not, the invariant diffuse boundary state has positive normalized diffuse oscillation mass, no unit-scale active observable, and non-affine velocity content. This is precisely the critical

diffuse-tail alternative D3. The alternatives are ordered by first extracting all active regeneration and then all affine/parasitic collapse; after those removals, the remaining nonzero diffuse state is $\mathcal{C}_{\text{critdiff}}$. $\square$

Definition 8.17 (Critical-tail compactification). The class $\mathcal{C}_{\text{critdiff}}$ consists of recurrent diffuse-defect states produced by [Theorem 8.16](#) which are supported on the boundary of the covariant observer compactification, carry normalized affine oscillation mass one, annihilate every fixed unit active observable, and are not affine/parasitic.

The critical-tail compactification $\mathfrak{T}_M$ is the compact state space of triples

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda)$$

generated by such critical diffuse states after macroscopic blow-down. Here $\bar{W}$ is the barycentric macroscopic velocity, $\mathcal{Y}_{x,s}$ is the Young measure recording unresolved bounded macroscopic oscillation, and Λ is the log-scale diffuse measure recording mass escaping every fixed multiplicative annulus. The coherent subspace $\mathfrak{T}_{\text{coh}}$ is the special case

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}, \quad \Lambda = 0.$$

The purely log-diffuse subspace is denoted $\mathfrak{T}_{\text{logdiff}}$, and the oscillatory Young-tail subspace is denoted $\mathfrak{T}_{\text{Young}}$.

For coherent elements, the corresponding macroscopic tangent pair (W, Q) belongs to $\mathcal{C}_{\text{coherit}}$ and satisfies on annular cylinders $\{0 < r_0 < |x| < r_1 < \infty\} \times I$

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \quad \nabla \cdot W = 0$$

in distributions, with nonzero affine-normalized diffuse oscillation and recurrence under the induced logarithmic dilation flow.

Theorem 8.18 (Critical-tail compactification extraction). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a bounded normalized terminal residual sequence, $\|U_n\|_{L^\infty} \leq M$. Assume that, after removal of all affine-normalized active covariant observers, there remains a nonzero diffuse exterior oscillation defect. Then, after passing to a subsequence, the sequence generates a nonzero element*

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M.$$

Moreover, the barycentric velocity satisfies the linear critical-tail transport law

$$\partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \quad \nabla \cdot \bar{W} = 0$$

in distributions on macroscopic annular cylinders, and the full defect $(\mathcal{Y}, \Lambda)$ is transported by the induced log-dilation flow.

Proof. Let μ_n^{osc} be the affine-normalized oscillation defect measure after active-locus removal. By the mixed-diffuse hypothesis there are escaping observer windows E_n and $m_0 > 0$ such that

$$\mu_n^{\text{osc}}(E_n) \geq m_0.$$

Normalize

$$\lambda_n := \frac{\mu_n^{\text{osc}} \lfloor E_n}{\mu_n^{\text{osc}}(E_n)}.$$

By [Theorem 8.13](#), after passing to a subsequence these probability measures converge to a boundary diffuse-defect state. Because all positive-threshold active observers have been removed, the limit gives no mass to ordinary unit active observers in Z .

We now compactify the critical boundary without imposing annular tightness. Apply Prokhorov compactness first in the observer compactification and then in the logarithmic radial compactification. If the normalized defect is tight in a fixed multiplicative annulus, select macroscopic scales

$R_n \rightarrow \infty$, observers z_n , affine gauges $c_n(s)$, and pressure gauges. On fixed annular cylinders set, schematically,

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s).$$

The uniform bound $\|U_n\|_{L^\infty} \leq M$ gives weak-* compactness of W_n in L_{loc}^∞ , and the fundamental theorem of Young measures gives, after passing to a further subsequence, a barycenter $\bar{W}$ and Young measure $\mathcal{Y}_{x,s}$. If the sequence is strongly compact in L_{loc}^3 , then $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$; otherwise $\mathcal{Y}$ records the unresolved oscillation defect. If the defect is not tight in any fixed logarithmic annulus, the escaping logarithmic mass is recorded by Λ . These alternatives are exhaustive and define an element of $\mathfrak{T}_M$. Since the normalized defect mass is one, $\mathcal{T} \neq 0$.

It remains to identify the equation satisfied by the barycenter. After the change $y = y_n + R_n x$ and the admissible affine/pressure normalizations, the centered equation becomes, in distributions on fixed macroscopic annuli,

$$\partial_s W_n - R_n^{-2} \Delta_x W_n + R_n^{-1} W_n \cdot \nabla_x W_n + \frac{1}{2} x \cdot \nabla_x W_n + \frac{1}{2} W_n + \nabla_x Q_n = o(1).$$

For every compactly supported test function, the viscous and nonlinear contributions are bounded respectively by

$$C R_n^{-2} \|W_n\|_{L^\infty}, \quad C R_n^{-1} \|W_n\|_{L^\infty}^2,$$

and hence both tend to zero. The pressure gauges give distributional compactness of the remaining gradients on annuli. Passing to the weak-* barycentric limit yields the displayed linear critical-tail transport equation. Applying the same weak formulation to transported test functions shows that $(\mathcal{Y}, \Lambda)$ is transported by the induced log-dilation flow. This construction is local on macroscopic annuli and uses no global critical norm, CKN budget, or energy estimate. $\square$

Corollary 8.19 (Coherent extraction under annular tightness and strong compactness). *If the critical-tail defect $\mathcal{T} \in \mathfrak{T}_M$ produced by Theorem 8.18 satisfies $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{W(x,s)}$ for a.e. (x, s) , then there exist macroscopic scales, covariant observers, affine gauges, and pressure gauges for which the blown-down fields converge strongly in L_{loc}^3 and weak-* in L_{loc}^∞ to a nonzero coherent critical tail $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$.*

Proof. This is the coherent subcase of Theorem 8.18. The condition $\Lambda = 0$ rules out loss of mass across logarithmic scale, and the Dirac-valued Young measure gives strong L_{loc}^3 convergence by uniform L^∞ boundedness. Nontriviality follows from the normalized affine oscillation mass. $\square$

Theorem 8.20 (Coherent critical-tail classification). *Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be a coherent critical tail. In logarithmic radial variables*

$$x = e^\rho \omega, \quad Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s),$$

its compact log-translation/time-translation hull belongs to exactly one of

$$\mathcal{C}_{\text{affine}}, \quad \mathcal{C}_{\text{homcrit}}, \quad \mathcal{C}_{\text{logper}}, \quad \mathcal{C}_{\text{ap}}.$$

Here $\mathcal{C}_{\text{homcrit}}$ is the homogeneous (-1) -tail class, $\mathcal{C}_{\text{logper}}$ is the nonzero log-periodic tail class, and $\mathcal{C}_{\text{ap}}$ is the remaining aperiodic minimal log-translation hull class.

Proof. Let

$$\mathcal{H}(Z) := \overline{\{Z(\rho + a, \omega, s + b) : a, b \in \mathbb{R}\}}$$

be the compact logarithmic translation hull. If the affine-normalized oscillation vanishes on every compact log-cylinder, then the tail is affine/parasitic. Otherwise, if every element of $\mathcal{H}(Z)$ is independent of ρ , then W is (-1) -homogeneous and belongs to $\mathcal{C}_{\text{homcrit}}$. If the log-translation orbit has a nonzero period, then W is discretely self-similar at infinity and belongs to $\mathcal{C}_{\text{logper}}$. If neither alternative holds, compact-flow decomposition gives a minimal compact invariant

subsystem with no nonzero log period; this is the aperiodic log-translation class $\mathcal{C}_{\text{ap}}$. These alternatives exhaust compact log-translation hulls. $\square$

Assumption 8.21 (Realized critical-tail rigidity). Every nonzero critical-tail defect $\mathcal{T} \in \mathfrak{T}_M$ generated by a bounded normalized residual Navier–Stokes profile satisfies one of the following alternatives:

- (i) $\mathcal{T}$ regenerates affine-normalized active covariant observers;
- (ii) $\mathcal{T}$ collapses to the affine/parasitic lower stratum;
- (iii) $\mathcal{T}$ lies in one of the structured critical-tail lower strata already removed before entering $\mathcal{C}_{\text{gen}}$:

$$\mathcal{C}_{\text{homcrit}}, \quad \mathcal{C}_{\text{logper}}, \quad \mathcal{C}_{\text{ap}}, \quad \mathcal{C}_{\text{Young}}, \quad \mathcal{C}_{\text{log diff}}.$$

Equivalently, the realized residual class contains no nonzero critical-tail defect in

$$\mathfrak{T}_{\text{coh}} \cup \mathfrak{T}_{\text{Young}} \cup \mathfrak{T}_{\text{log diff}}$$

after active regeneration, affine/parasitic collapse, and the listed structured critical-tail strata have been removed.

Theorem 8.22 (No mixed compact–diffuse residual state). *Assume the realized critical-tail rigidity input Theorem 8.21. Let (U, Π) be a retained terminal profile in the refined residual class, after the affine normalization of Theorem 7.12. Then U cannot have a diffuse exterior oscillation remainder, nor an escaping retained noncompact recentering remainder, coexisting with its compact retained core. Equivalently, the paired terminal states*

$$\text{one active compact core} + \mathcal{R}_{\text{diff}}, \quad \text{one active compact core} + \mathcal{R}_{\text{noncomp}},$$

and their finite, infinite, or continuum active-locus analogues do not occur in the refined residual class.

Proof. The noncompact escaping case is local. In this theorem the noncompact remainder is retained in the velocity or affine-oscillation sense. If an escaping recentering with retained mass loses the local dissipation or pressure-gauge bounds in Theorem 8.3, then after its own local normalization it is a retained compact-mass recentering sequence. This contradicts Theorem 8.5. Thus it remains to exclude a compact core plus diffuse exterior remainder.

Assume that such a diffuse exterior remainder exists. By the paired active-locus decomposition Theorem 8.9, remove neighborhoods of the active locus at every positive affine-normalized activity threshold. The mixed assumption says that a nonzero diffuse oscillation residual remains. Apply Theorem 8.16.

If active regeneration occurs, then a positive oscillation-active observer survives after all positive-threshold active loci were removed, contradicting the definition of the diffuse residual. If affine/parasitic collapse occurs, then Theorems 7.12 and 9.10 places the relevant descendant hull in an already closed lower stratum. This cannot occur inside $\mathcal{C}_{\text{gen}}$.

The only remaining alternative is a critical diffuse tail. By Theorem 8.18, it generates a nonzero $\mathcal{T} \in \mathfrak{T}_M$. Apply Theorem 8.21. Active regeneration and affine/parasitic collapse have already been excluded, and every structured critical-tail lower stratum listed in the remaining alternative has been removed before $\mathcal{C}_{\text{gen}}$ is entered. All alternatives are impossible. Hence no mixed compact–diffuse residual profile exists. $\square$

Definition 8.23 (Finite separated family of retained concentration profiles). A finite separated family of retained concentration profiles is a finite family of recentering sequences. For the original terminal extraction these are the parabolic recenterings of Theorem 8.1; for descendants of a centered profile they are the covariant tail recenterings of Theorem 7.4. We write them uniformly as $g_{1,n}, \dots, g_{J,n}$, $J \geq 2$, such that:

- (i) the recentering sequences are mutually separated;
- (ii) each recentered sequence converges locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo gauges to a terminal profile (U_j, Π_j) ;
- (iii) each limiting profile has retained local concentration:

$$\mathcal{E}_1(U_j, \Pi_j; 0) \geq \varepsilon_* \quad (j = 1, \dots, J).$$

This is a finite local family, not a statement that all concentration profiles in the whole space have been counted. The separated-profile proof below treats the possible continuation of escaping recenterings by local concentration alternatives, not by a global counting estimate.

Proposition 8.24 (Finite separated families reduce to single-profile alternatives). *Assume the single-profile retained concentration closure statement: no single retained concentration profile remains in the refined residual decomposition after the already closed local classes are removed. Then a finite separated family is not a new refined residual alternative.*

Proof. Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$ be a finite separated concentrating family, represented by mutually separated parabolic recentering sequences. Recenter the terminal sequence at the sequence defining $\mathfrak{s}_j$. In that frame all other concentration profiles leave every compact terminal cylinder, so the local terminal limit is a single retained concentration profile. By the single-profile closure statement, this component either belongs to the already closed local classes, has local vanishing, or has loss of local compactness. Since the original family has retained concentration and local compactness, the latter two alternatives are incompatible with the hypotheses. Applying this argument to each j shows that the finite separated family consists only of already closed local profiles, and hence does not remain in the refined residual decomposition. $\square$

9. SEPARATED CONCENTRATION PROFILES, INDECOMPOSABILITY, AND SEQUENCE- L^3

We now close the terminal alternatives carrying retained local concentration. The proof uses no global critical-mass budget. Instead, finite separated concentration families are treated through local recentering and compactness. Iterating retained tail limits from such a finite family has only local state-space outcomes: an indecomposable retained concentration profile, a mixed compact–diffuse or compact–noncompact successor, or an active descendant branch. The mixed successors are excluded by [Theorem 8.22](#); infinite active branches and finite loops are excluded by the compact active descendant theorem.

Definition 9.1 (Tail limits and terminal indecomposability). Let (U, Π) be a terminal profile with retained local concentration. A retained concentrating tail limit is a limit of covariant tail recenterings

$$\mathcal{T}_{(x_n, \tau_n)}(U, \Pi), \quad |x_n| + |\tau_n|^{1/2} \rightarrow \infty,$$

in the local smooth velocity topology and the local $L^{3/2}$ pressure topology modulo gauges, with retained local concentration at the origin. A raw fixed spatial shift is not used in this definition; by [Theorems 7.3](#) and [7.4](#), it would introduce an unbounded spurious drift for escaping centers. Diffuse exterior concentration means that there are $\tau_n \rightarrow -\infty$, $R_n \rightarrow \infty$, and $\eta > 0$, such that

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq \eta,$$

while every unit-scale exterior covariant recentering sequence remains below the retained-concentration threshold, unless that recentering sequence has loss of local compactness. An escaping non-compact recentering sequence is a covariant tail recentering sequence for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails. A retained concentration profile is terminally indecomposable if it has no retained concentrating tail limit, no diffuse exterior concentration, and no escaping non-compact recentering sequence.

Lemma 9.2 (Time-slice activity gives parabolic CKN activity). *Let (U, Π) be a bounded smooth solution of (1.2) with $\|U\|_{L^\infty} \leq M$. If*

$$\int_{B_1(0)} |U(y, 0)|^3 dy \geq \varepsilon_0 > 0,$$

then there are $\delta = \delta(M, \varepsilon_0) > 0$ and $c_0 = c_0(M, \varepsilon_0) > 0$ such that, after the standard local pressure gauge,

$$\iint_{B_1(0) \times (-\delta^2, 0)} |U|^3 dy d\tau \geq c_0, \quad \text{and hence} \quad \mathcal{E}_1(U, \Pi; 0) \geq c_0.$$

The same constants apply to every covariant tail recentering $\mathcal{T}_{(x, \sigma)}(U, \Pi)$.

Proof. Apply interior parabolic regularity on $B_2 \times [-2\delta_0, \delta_0]$, with $\delta_0 > 0$ universal and small. On this cylinder the drift $-\frac{1}{2}y \cdot \nabla U$ has bounded smooth coefficients, and the nonlinear forcing is bounded by M^2 . After choosing local pressure gauges, the standard interior estimates give

$$\|\partial_\tau U\|_{L^\infty(B_1 \times [-\delta_0, 0])} + \|\nabla U\|_{L^\infty(B_1 \times [-\delta_0, 0])} \leq C(M)$$

where $C(M)$ is independent of the particular profile. Hence the map $\tau \mapsto \int_{B_1} |U(y, \tau)|^3 dy$ is uniformly continuous with a modulus depending only on M and $|B_1|$. Choose $\delta \leq \delta_0$ so that

$$\int_{B_1} |U(y, \tau)|^3 dy \geq \frac{1}{2}\varepsilon_0 \quad -\delta^2 < \tau < 0.$$

Integrating in time gives a positive lower bound for the velocity part of $\mathcal{E}_1(U, \Pi; 0)$; the pressure contribution is nonnegative after any admissible gauge. The constants are unchanged under $\mathcal{T}_{(x, \sigma)}$, because that recentering is an exact composition of the centered symmetries in Theorem 7.4. $\square$

Lemma 9.3 (One retained tail limit gives two separated concentration profiles). *Let U be a retained concentration terminal profile. If U has a retained concentrating tail limit, then the corresponding terminal extraction contains a finite separated family of retained concentration profiles.*

Proof. The origin recentering sequence carries retained concentration:

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Let W be a retained concentrating tail limit. By definition there are covariant relative tail recentering parameters $\zeta_n = (x_n, \tau_n)$, with $|x_n| + |\tau_n|^{1/2} \rightarrow \infty$, such that $\mathcal{T}_{\zeta_n}(U, \Pi) \rightarrow (W, \Pi_W)$ in the terminal topology and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

By local convergence of velocity and pressure after gauges,

$$\mathcal{E}_1(\mathcal{T}_{\zeta_n} U, \mathcal{T}_{\zeta_n} \Pi; 0) \geq \frac{1}{2}\varepsilon_*$$

for all sufficiently large n . Thus, for large n , the origin recentering sequence and the covariant tail sequence ζ_n are separated and have retained local concentration. They form a finite separated two-profile concentration family. $\square$

Proposition 9.4 (No finite separated family and local exclusions imply indecomposability). *Every single retained concentration terminal profile arising from a refined residual candidate is terminally indecomposable.*

Proof. Let U be a single retained concentration terminal profile. If U had a retained concentrating tail limit, [Theorem 9.3](#) would produce a finite two-profile separated family with retained concentration, contradicting [Theorem 9.19](#). If U had diffuse exterior concentration or an escaping non-compact recentering sequence, the corresponding terminal extraction would be a mixed state consisting of the retained compact core U plus a diffuse or noncompact tail remainder. This is excluded by [Theorem 8.22](#). Hence no retained concentrating tail limit, diffuse exterior concentration, or escaping non-compact recentering sequence exists, and U is terminally indecomposable. $\square$

Lemma 9.5 (Indecomposable sequence- L^3 completeness). *Let (U, Π) be a bounded smooth ancient solution of (1.2). Suppose U has retained local concentration and is terminally indecomposable in the sense of [Theorem 9.1](#). Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We prove the contrapositive. Define

$$A(\tau) := \int_{\mathbb{R}^3} |U(y, \tau)|^3 dy \in [0, \infty].$$

If the conclusion fails, then

$$\liminf_{\tau \rightarrow -\infty} A(\tau) = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $A(\tau_n) \rightarrow \infty$, allowing the value $+\infty$. Since U is bounded by some M , for each finite R ,

$$\int_{|y| \leq R} |U(y, \tau_n)|^3 dy \leq M^3 |B_R|.$$

Choose $R_n \rightarrow \infty$ so slowly that, after discarding finitely many n ,

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

If $A(\tau_n) = +\infty$, take $R_n = n$; otherwise choose R_n with $M^3 |B_{R_n}| \leq A(\tau_n)/2$.

Set

$$m_n := \sup_{|x| > R_n} \int_{B_1(x)} |U(y, \tau_n)|^3 dy.$$

There are two alternatives. First suppose $\limsup_{n \rightarrow \infty} m_n \geq \varepsilon_*$. Then, passing to a subsequence, there are x_n with $|x_n| > R_n \rightarrow \infty$ and

$$\int_{B_1(x_n)} |U(y, \tau_n)|^3 dy \geq \varepsilon_*.$$

Use the covariant tail recentering

$$U_n(y, s) := U(y + e^{s/2} x_n, \tau_n + s).$$

At $s = 0$,

$$\int_{B_1(0)} |U_n(y, 0)|^3 dy \geq \varepsilon_*.$$

If this recentering sequence has loss of local compactness, this contradicts terminal indecomposability. If it has local compactness, the compactness theorem gives a subsequential smooth limit W , and

$$\int_{B_1(0)} |W(y, 0)|^3 dy \geq \varepsilon_*,$$

after lowering ε_* by a harmless fixed factor if necessary. By [Theorem 9.2](#), this time-slice velocity concentration gives parabolic CKN activity on $Q_1(0,0)$. Thus W is a retained concentrating covariant tail limit, again contradicting terminal indecomposability.

The remaining alternative is that $m_n < \varepsilon_*$ for all sufficiently large n . Together with the exterior lower bound, this says that at times τ_n there is at least one unit of L^3 -mass outside B_{R_n} , but no unit-scale exterior recentering sequence carries retained local concentration. If some such exterior recentering sequence has loss of local compactness, terminal indecomposability is contradicted. If no such recentering sequence has loss of local compactness, this is diffuse exterior concentration with $\eta = 1$, again contradicting terminal indecomposability. Both alternatives are impossible, so the desired sequence exists. $\square$

Theorem 9.6 (Parasitic-free mildness inheritance from the normalized package). *Assume the Paper I normalized Seregin extraction package together with the Paper II bounded mild centered normalization. Let (U, Π) be a retained terminal profile, retained covariant tail descendant, active-tree limit, or recurrent-core limit represented in the normalized residual state space. Then exactly one of the following alternatives applies.*

- (i) (U, Π) belongs to the affine/parasitic lower stratum of [Theorem 7.12](#);
- (ii) after the canonical affine normalization already built into the residual representative, the physical pullback

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0,$$

is parasitic-free and has the following property: for every $T < 0$, the finite-terminal shift

$$u^T(x, s) := u(x, T + s), \quad s < 0,$$

is a bounded mild ancient Navier–Stokes solution in the Albritton–Barker sense on $\mathbb{R}^3 \times (-\infty, 0)$.

The dichotomy and the normalized mildness property are part of this imported normalization convention and are preserved by the local state-space operations: Seregin extraction, centered time shifts, covariant translations, covariant tail descendants, active-tree limits, and recurrent-core limits.

Proof. Affine-constant centered profiles are precisely the homogeneous parasitic profiles removed by [Theorem 7.12](#); they are assigned to alternative (i). We therefore consider a representative outside that lower stratum. The Paper I normalization, interpreted through the bounded mild centered formulation of Paper II, is imported as part of the normalized residual representative: the spatially homogeneous parasitic mode has already been removed before the residual terminal analysis begins. The local state-space operations T_a , Θ_s , $\mathcal{T}_{(x,\sigma)}$, and S_c act by exact centered symmetries and admissible gauge changes, with the S_c bound tracked as in [Theorem 7.9](#). Hence they carry normalized representatives to normalized representatives, unless the limit is affine/parasitic, in which case the ordered decomposition assigns it to the affine lower stratum. This is bookkeeping for the imported normalization; it is not a new global estimate.

Fix $T < 0$. On $\mathbb{R}^3 \times (-\infty, T]$, the physical pullback is smooth and bounded by $(-T)^{-1/2} \|U\|_{L^\infty}$, and it solves the physical Navier–Stokes equations by the direct self-similar change of variables. Since the parasitic zero mode has been removed, the pressure has no unprojected linear parasitic component on this finite terminal interval. Hence the bounded mild formulation of Paper II applies: for $-\infty < s < t \leq T$,

$$u(t) = e^{(t-s)\Delta} u(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(\sigma) d\sigma.$$

After translating the terminal time T to 0, this is the mild ancient formulation for u^T on every finite forward interval $(-\infty, 0)$. The bound $\|u^T\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))} \leq (-T)^{-1/2} \|U\|_{L^\infty}$ is finite. Thus u^T is a genuine bounded mild ancient solution in the sense required by the Albritton–Barker endpoint theorem. $\square$

Theorem 9.7 (Albritton–Barker endpoint Liouville theorem). *Let u be a mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists $t_k \downarrow -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem in the sequence- L^3 form proved by Albritton and Barker [5]. Paper I records the same peer-reviewed input among its auxiliary Liouville results [2]. $\square$

Lemma 9.8 (No retained indecomposable concentration profile). *No retained terminally indecomposable concentration profile exists.*

Proof. Let U have retained local concentration and be terminally indecomposable. By Theorem 9.5, choose $\tau_k \rightarrow -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let u be the physical pullback and set $t_k = -e^{-\tau_k}$. Since $\tau_k \rightarrow -\infty$, the sequence satisfies $t_k \downarrow -\infty$, and critical scaling gives

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By Theorem 9.6, either U is in the affine/parasitic lower stratum, which is not part of the normalized residual class, or else for each $T < 0$

$$u^T(x, s) := u(x, T + s), \quad s < 0,$$

is a bounded mild ancient solution. Choose $T < 0$ and discard finitely many indices so that $t_k < T$. Then $s_k := t_k - T \downarrow -\infty$, and

$$\|u^T(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} = \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} \leq \sup_j \|U(\cdot, \tau_j)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Theorem 9.7 gives $u^T \equiv 0$. Since $T < 0$ was arbitrary, the physical pullback vanishes on $\mathbb{R}^3 \times (-\infty, 0)$, and therefore $U \equiv 0$. This contradicts retained local concentration. $\square$

Lemma 9.9 (Diagonal lifting of retained concentrating tail limits). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate, represented by separated recentering sequences $g_{j,n} = (y_{j,n}, \tau_{j,n})$. Let (U_j, Π_j) be the terminal profile obtained in the j -th recentering. If (U_j, Π_j) has a retained concentrating tail limit (W, Π_W) , then there is a combined raw-covariant descendant recentering sequence $\mathfrak{d} = (d_n)$, in the sense of Theorem 8.2, for the original terminal sequence such that:*

- (i) $\mathfrak{d}$ is asymptotically separated from g_j ;
- (ii) the recentered original sequence at $\mathfrak{d}$ converges locally to (W, Π_W) , after admissible pressure gauges;
- (iii) $\mathfrak{d}$ has retained local concentration.

Proof. By definition of retained concentrating tail limit, there are relative covariant tail recenterings $\zeta_m = (\xi_m, \sigma_m)$, with

$$|\xi_m| + |\sigma_m|^{1/2} \rightarrow \infty,$$

such that

$$\mathcal{T}_{\zeta_m}(U_j, \Pi_j) \rightarrow (W, \Pi_W)$$

locally in the terminal topology, and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Fix m . Since $V_n^{g_j} \rightarrow U_j$ locally smoothly and $(P_n^{g_j}) \rightarrow \Pi_j$ locally in $L^{3/2}$ modulo gauges, the convergence also holds on the fixed covariant tube

$$\{(y + e^{s/2}\xi_m, s + \sigma_m) : (y, s) \in Q_m(0, 0)\}.$$

Therefore one can choose $n(m)$ so large that the covariantly recentered original fields

$$V_{n(m)}(y + y_{j,n(m)} + e^{s/2}\xi_m, s + \tau_{j,n(m)} + \sigma_m)$$

and the corresponding pressure gauges are within m^{-1} of the $\mathcal{T}_{\zeta_m}$ -recentered fields of (U_j, Π_j) on $Q_m(0, 0)$.

Set

$$d_m := (g_{j,n(m)}; \xi_m, \sigma_m).$$

Relabeling the subsequence $n(m)$ as the terminal index gives a combined raw-covariant descendant recentering sequence $\mathfrak{d} = (d_m)$ for the original terminal sequence. Since the covariant parabolic distance from d_m to $g_{j,n(m)}$ is $|\xi_m| + |\sigma_m|^{1/2}$, the sequence $\mathfrak{d}$ is separated from the j -th recentering sequence. The diagonal choice gives local convergence of the original sequence at $\mathfrak{d}$ to (W, Π_W) . Let $\mu_{n(m)}^\mathfrak{d}$ denote the local CKN measure of the original terminal sequence pulled back by this combined raw-covariant recentering and by the pressure gauges used in the diagonal convergence. The local CKN quantity is continuous under the strong local velocity convergence and weak lower semicontinuity of the pressure term after these fixed gauges, so

$$\liminf_{m \rightarrow \infty} \mu_{n(m)}^\mathfrak{d}(Q_1(0, 0)) \geq \mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Thus $\mathfrak{d}$ has retained local concentration. □

Theorem 9.10 (Heredity of retained descendants). *Let (U, Π) be a normalized retained residual profile in $\mathfrak{X}_M$, and let (W, Π_W) be a retained covariant tail descendant of (U, Π) . Then $(W, \Pi_W) \in \mathfrak{X}_M$, it carries retained local concentration at the descendant threshold fixed in [Theorem 2.2](#), and it has the same parasitic-free normalization status as a terminal profile generated directly by the [Paper I](#) extraction. Consequently exactly one of the following ordered outcomes occurs:*

- (i) (W, Π_W) belongs to one of the previously closed lower strata, in which case that descendant is already discharged;
- (ii) (W, Π_W) belongs to the normalized residual terminal state space, so the active-tree and terminal-indivisibility arguments may be iterated from W .

Proof. The descendant is obtained as a limit of covariant tail recenterings $\mathcal{T}_{\zeta_n}(U, \Pi)$. By [Theorem 7.9](#), the recentered sequence is locally precompact in $\mathfrak{X}_M$, and the limit remains a bounded smooth locally suitable centered solution with the same L^∞ bound and compatible pressure gauges. Retained local concentration is part of the definition of retained descendant; when the descendant is produced through a limit passage, [Theorem 8.4](#) fixes the threshold loss once and for all through [Theorem 2.2](#). The affine/parasitic normalization is canonical for the residual representative and is preserved by T_a , Θ_s , and their composition $\mathcal{T}_{\zeta_n}$; if a limit is affine-parasitic, it is assigned to the affine-constant lower stratum of [Theorem 7.12](#). Otherwise it is represented in the same normalized residual state space. The ordered residual decomposition then places the descendant either in a previously closed lower class or in the generic residual state space. This is local state-space bookkeeping and does not use a global critical norm or global tail budget. □

Lemma 9.11 (Escaping recenterings in a separated family). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. For each component $\mathfrak{s}_j$, the following ordered alternatives exhaust the possible successor states:*

- (i) *the profile (U_j, Π_j) is terminally indecomposable;*
- (ii) *(U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness;*
- (iii) *(U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.*

Proof. This is the definition of terminal indecomposability plus the diagonal lifting lemma, with no disjointness assertion. If (U_j, Π_j) is terminally indecomposable, (i) holds. If it is not terminally indecomposable, at least one of the three excluded alternatives in [Theorem 9.1](#) occurs: a retained concentrating tail limit, diffuse exterior concentration, or an escaping recentering sequence with loss of local compactness. The diffuse exterior and local-compactness-failure cases give (ii). In the retained concentrating tail-limit case, [Theorem 9.9](#) lifts the limit to a recentering sequence of the original terminal sequence with retained local concentration, and [Theorem 9.10](#) places the descendant either in a closed lower stratum or back in the normalized residual state space, giving (iii) in the iterative residual case. If several successor states coexist, they are all recorded; the next lemma removes the nonactive ones. $\square$

Lemma 9.12 (No diffuse or non-compact escaping recentering from a retained family). *In the setting of [Theorem 9.11](#), alternative (ii) cannot occur.*

Proof. Suppose that (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness. By the same diagonal procedure used in [Theorem 9.9](#), this tail state is realized by a terminal extraction of the original refined residual candidate, after passing to a subsequence and choosing the local pressure gauges supplied by Paper I. It is not enough to say that the diffuse or noncompact state is not the sole carrier of the original compact concentration: the compact core (U_j, Π_j) may coexist with that tail state. The paired exclusion [Theorem 8.22](#) rules out exactly this coexistence in the refined residual class. Therefore alternative (ii) cannot occur for a refined residual family with retained local concentration. $\square$

Definition 9.13 (Closed active successor relation). For a fixed threshold $\eta > 0$, let

$$\mathfrak{A}_\eta := \{U \in \mathfrak{X}_M : \Phi(U) \geq \eta\}$$

be the closed affine-normalized activity set of [Theorem 7.14](#). We write

$$U \mathcal{R}_\eta V$$

if $U, V \in \mathfrak{A}_\eta$ and V is a retained active covariant tail descendant of U .

Lemma 9.14 (Closedness of the active successor relation). *The relation*

$$\mathcal{R}_\eta \subset \mathfrak{A}_\eta \times \mathfrak{A}_\eta$$

is closed in the local $\mathfrak{X}_M$ topology.

Proof. Let $U_m \rightarrow U$, $V_m \rightarrow V$ in $\mathfrak{X}_M$, with $U_m \mathcal{R}_\eta V_m$. For each m , choose a retained covariant tail observer sequence $\zeta_{m,\ell}$ for U_m whose recentered fields converge to V_m on $Q_m(0,0)$ to accuracy m^{-1} , after admissible pressure gauges. A diagonal choice in (m, ℓ) , using [Theorems 7.9](#) and [8.4](#), gives a single escaping covariant tail sequence for U converging locally to V . Closedness of $\mathfrak{A}_\eta$ follows from [Theorem 7.14](#), and lower semicontinuity of the retained local CKN quantity after the fixed gauges preserves the descendant threshold. Hence $U \mathcal{R}_\eta V$. $\square$

Theorem 9.15 (Active path-space recurrence). *Assume there exists an infinite active descendant chain*

$$U_0 \mathcal{R}_\eta U_1 \mathcal{R}_\eta U_2 \mathcal{R}_\eta \cdots$$

Then the path space

$$\mathcal{P}_\eta := \left\{ (W_j)_{j \geq 0} \in \mathfrak{A}_\eta^\mathbb{N} : W_j \mathcal{R}_\eta W_{j+1} \text{ for all } j \right\}$$

is nonempty and compact, is invariant under the left shift

$$S(W_0, W_1, W_2, \dots) = (W_1, W_2, \dots),$$

and carries an S -invariant Borel probability measure $\mathbb{P}$. Its projected stationary state measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported entirely on $\mathfrak{A}_\eta$.

Proof. The set $\mathfrak{A}_\eta$ is compact in the local state topology inherited from the bounded Seregín state space. Hence $\mathfrak{A}_\eta^\mathbb{N}$ is compact and metrizable. By [Theorem 9.14](#),

$$\mathcal{P}_\eta = \bigcap_{j=0}^{\infty} \{(W_k)_{k \geq 0} : (W_j, W_{j+1}) \in \mathcal{R}_\eta\}$$

is closed, hence compact. The given infinite chain is an element of $\mathcal{P}_\eta$, so the path space is nonempty, and it is plainly invariant under the left shift.

For the path $p = (U_0, U_1, \dots)$, define empirical measures

$$\mathbb{P}_N := \frac{1}{N} \sum_{k=0}^{N-1} \delta_{S^k p}.$$

Weak-* compactness of probability measures on compact $\mathcal{P}_\eta$ gives a convergent subsequence $\mathbb{P}_{N_\ell} \rightharpoonup \mathbb{P}$. For each continuous F on $\mathcal{P}_\eta$,

$$\int F \circ S d\mathbb{P}_N - \int F d\mathbb{P}_N = \frac{F(S^N p) - F(p)}{N} \rightarrow 0.$$

Thus $\mathbb{P}$ is shift-invariant. Since every coordinate of every path in $\mathcal{P}_\eta$ belongs to $\mathfrak{A}_\eta$, the projected measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported in $\mathfrak{A}_\eta$. $\square$

Theorem 9.16 (No infinite active descendant branch). *No infinite retained active descendant branch can occur in the refined residual class.*

Proof. Suppose an infinite retained active branch exists. If $\limsup_j \Phi(U_j) = 0$, local compactness gives a subsequential limit $U_{j_k} \rightarrow U_\infty$ with $\Phi(U_\infty) = 0$. The diagonal lifting used in [Theorem 9.9](#) shows that this limit is still a retained tail descendant of the original residual profile. By [Theorem 7.15](#), U_∞ belongs to the affine/parasitic lower stratum, contradicting descendant heredity in the refined residual class. Hence $\limsup_j \Phi(U_j) > 0$. Choose $\eta > 0$ and an infinite ordered subbranch contained in $\mathfrak{A}_\eta$. By transitivity of retained tail descendants through [Theorem 9.9](#), this ordered subbranch is an $\mathcal{R}_\eta$ -chain.

[Theorem 9.15](#) gives a compact shift-invariant active path space and a stationary path measure supported on $\mathfrak{A}_\eta$. By Poincaré recurrence for the shift, a recurrent path returns arbitrarily close to itself. Concatenating the retained tail-limit observers along the returning blocks and using the diagonal lifting lemma produces a compact tail-minimal recurrent core for (1.2) whose states lie in $\mathfrak{A}_\eta$. The recurrent-core rigidity theorem [Theorem 7.22](#) forces this core to be affine-constant after the canonical affine normalization. But affine/parasitic profiles have $\Phi = 0$, whereas $\mathfrak{A}_\eta = \{\Phi \geq \eta\}$. This contradiction rules out the infinite active branch. $\square$

Lemma 9.17 (Finite separated families produce indecomposable profiles or active branches). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. Then either one of the terminal profiles (U_j, Π_j) is terminally indecomposable, or the family generates an infinite active descendant branch. A finite recurrent loop is counted as an infinite branch by periodic repetition.*

Proof. Assume no (U_j, Π_j) is terminally indecomposable. By Theorems 9.11 and 9.12, each (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.

If a lifted concentrating recentering sequence is not equivalent to one of the sequences already in the finite family, enlarge the family by adding this new sequence. This is still a finite retained concentration separated family because only one separated recentering sequence with retained local concentration has been added. Repeat the construction. At each finite stage there are two possibilities. Either an indecomposable profile appears, proving the first alternative, or every existing decomposable profile has a retained concentrating tail limit.

If the process returns to a previously occurring concentrating recentering sequence, the directed recentering relation contains a finite loop

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \dots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}.$$

Repeating the loop periodically gives an infinite active descendant branch. If the process never returns to a previously occurring concentrating class, the successively lifted retained tail limits themselves form an infinite active descendant branch. The nonactive successor alternatives at each stage are excluded by Theorem 9.12, while coexistence of a compact core with a diffuse or noncompact remainder is handled separately by Theorem 8.22. $\square$

Lemma 9.18 (Finite recurrent loops of retained tail limits are impossible). *No refined residual terminal extraction contains a finite recurrent loop of retained concentrating tail limits.*

Proof. Let

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \dots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}$$

be such a loop, and write (U_ℓ, Π_ℓ) for the corresponding retained concentration terminal profiles. Each arrow means that $U_{\ell+1}$ is a retained concentrating tail limit of U_ℓ . Thus, for each ℓ , there are escaping tail recentering sequences whose recentered fields converge locally to $U_{\ell+1}$.

Let $\mathcal{K}$ be the closure, in the local smooth velocity topology and local $L^{3/2}$ pressure topology modulo gauges, of all profiles obtained from the finite set $\{U_0, \dots, U_{q-1}\}$ by time translations and covariant terminal translations. The boundedness inherited from the Type I Seregin extraction theorem and the local pressure compactness of Paper I give compactness of $\mathcal{K}$ on every fixed terminal cylinder; a diagonal argument gives compactness of $\mathcal{K}$. The set is invariant under the same time and covariant translations by construction. The recurrent-loop relations imply that each U_ℓ returns to $\mathcal{K}$ through tail limits, so the closed invariant subset of $\mathcal{K}$ generated by the loop is nonempty, compact, invariant, and contains the retained concentration profiles.

Repeating the loop periodically gives an infinite active descendant branch. This contradicts Theorem 9.16. Therefore no finite recurrent loop of retained concentrating tail limits exists. $\square$

Theorem 9.19 (Finite separated retained concentration profile families are impossible). *No refined residual candidate admits a finite separated family of retained concentration profiles.*

Proof. Suppose, for contradiction, that a refined residual candidate admits a finite retained concentration separated family $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$. By Theorem 9.17, either some component of the family is terminally indecomposable or the family generates an infinite active descendant branch. The first alternative contradicts Theorem 9.8. The second contradicts Theorem 9.16. Hence the finite retained concentration separated family cannot exist. $\square$

Theorem 9.20 (Terminal local concentration theorem). *Assume the realized critical-tail rigidity input Theorem 8.21. Every bounded centered Type I Seregin profile with retained local concentration outside the previously closed lower classes admits the local terminal concentration decomposition of Theorem 8.9 as an active retained profile locus plus a residual remainder. Pure local vanishing, exterior escape, and noncompact loss cannot be the sole locations of the distinguished compact mass by Theorem 8.12; mixed compact–diffuse and compact–noncompact states are excluded by Theorem 8.22; infinite active descendant branches are excluded by Theorem 9.16; finite separated retained concentration profile families are excluded by Theorem 9.19; every remaining single retained concentration terminal profile is terminally indecomposable by Theorem 9.4; retained indecomposable profiles have a backward sequence of finite critical norm by Theorem 9.5; and retained concentration terminal profiles are parasitic-free mild, unless they are already in the affine lower stratum, by Theorem 9.6.*

Proof. This is the assembly of the local results and explicit state-space inputs in Sections 8 and 9. The local concentration decomposition is the paired active-locus/remainder decomposition Theorem 8.9. The old pure radiative alternatives are only sole-carrier exclusions, and are used only in that form through Theorem 8.12. Coexistence of a compact active core with diffuse or noncompact tail remainder is separately ruled out by Theorem 8.22, using realized critical-tail rigidity Theorem 8.21. Infinite active trees and finite recurrent loops are ruled out by the active path-space recurrence theorem Theorem 9.16, while finite separated active families are ruled out by Theorem 9.19. Therefore any retained concentration terminal profile that survives the ordered lower-class removals is single and has no retained active descendant or diffuse/noncompact tail remainder. By Theorem 9.4 it is terminally indecomposable, and then Theorems 9.5 and 9.6 give the sequence- L^3 and parasitic-free finite-shift mildness conclusions. All estimates used in this assembly are localized to fixed terminal cylinders, except for the final sequence L^3 conclusion, which is derived by contradiction from terminal indecomposability and is not assumed as a global bound. $\square$

10. CLOSURE OF THE GENERIC TERMINAL RESIDUAL

Definition 10.1 (Generic residual profile). A generic residual profile is a smooth bounded normalized Seregin ancient solution of (1.2) which carries persistent compact velocity L^3 -mass and belongs to none of the previously closed alternatives, none of $\mathcal{C}_{\text{ax}}, \mathcal{C}_{\text{rot}}, \mathcal{C}_{\text{stat}}^{\text{ret}}$, and none of the structured or tight classes removed earlier.

Theorem 10.2 (Closure of the generic residual class by local concentration analysis). *Assume the realized critical-tail rigidity input Theorem 8.21. One has*

$$\mathcal{C}_{\text{gen}} = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{gen}}$. By the tail geometry developed above, constant tails may be removed by the exact affine symmetry and nonconstant recurrent profile tail cores collapse to affine constants. Thus the remaining representative may be taken to be tail-normalized without leaving the centered equation or the residual decomposition. If this normalization eliminates non-affine activity, then Theorem 7.15 places the profile in the affine/parasitic lower stratum already removed from $\mathcal{C}_{\text{gen}}$. Thus, for the residual representative still under consideration, a positive non-affine activity threshold survives.

Apply the paired terminal local concentration decomposition to a terminal extraction of this tail-normalized candidate. The distinguished compact velocity mass from Paper I still forces the terminal active locus to be nonempty by Theorem 8.10; after affine normalization the surviving residual activity is measured by Φ . Pure radiative alternatives cannot be the sole location of the compact mass by Theorem 8.12, and mixed compact–diffuse or compact–noncompact remainders

are excluded by [Theorem 8.22](#). Infinite active branches are excluded by [Theorem 9.16](#). Thus the active locus reduces, in the refined residual class, either to a finite separated family or to a single retained profile with no surviving tail remainder.

The finite separated-family case contradicts [Theorem 9.19](#). In the single-profile case, [Theorem 9.4](#) makes the terminal profile terminally indecomposable. Then [Theorem 9.8](#) excludes it. All paired state-space possibilities are impossible. Hence no generic residual profile exists. $\square$

11. FINAL ASSEMBLY

Assuming the realized critical-tail rigidity input [Theorem 8.21](#), the class closures above form the following dependency chain. The rotational relative-equilibrium arrow is the localized Coriolis-flux argument; the terminal active-locus compactness, no finite separated-family, and indecomposability arrows are proved in this paper from local estimates:

$$\begin{aligned} \mathcal{C}_{\text{ax}} &\xrightarrow{\text{r-Liouville}} \emptyset, \\ \mathcal{C}_{\text{rot}} &\xrightarrow{\text{local flux}} \emptyset, \\ \mathcal{C}_{\text{stat}}^{\text{ret}} &\xrightarrow{\text{terminal concentration}} \emptyset, \\ \mathcal{C}_{\text{gen}} &\xrightarrow{\text{critical-tail closure, finite separation, sequence-}L^3} \emptyset. \end{aligned}$$

The positive local velocity mass inherited by every normalized Seregin limit, after affine normalization through the non-affine activity functional Φ , is the common contradiction target in all four classes.

Equivalently, the final exclusion is the composition of five certificates:

- (1) *Entry certificate*: Paper 0 and Paper I turn a finite-energy local pointwise Type I singular point into a normalized Seregin profile with retained compact velocity concentration.
- (2) *Stratification certificate*: the ordered residual decomposition places every remaining retained profile in one of the refined strata $\mathcal{C}_{\text{ax}}, \mathcal{C}_{\text{rot}}, \mathcal{C}_{\text{stat}}^{\text{ret}}, \mathcal{C}_{\text{gen}}$.
- (3) *Closed-strata certificate*: the small, stationary L^3 , tight, closed structured/decay, axisymmetric, rotational, and stationary-hull strata are discharged by the results cited above.
- (4) *Residual terminal certificate*: the generic residual is reduced by the paired active-locus decomposition, diffuse-defect compactness, diffuse-defect trichotomy, critical-tail compactification, realized critical-tail rigidity, descendant heredity, active path-space recurrence, finite separated-family exclusion, and terminal indecomposability.
- (5) *Endpoint Liouville certificate*: a terminally indecomposable retained profile is either affine/parasitic lower-stratum or has a parasitic-free bounded mild physical pullback with a backward L^3 -bounded sequence, and is therefore zero by Albritton–Barker.

Proof of [Theorem 1.7](#), revisited. The detailed reductions have now been given in the four class sections. The axisymmetric bounded-circulation class is empty unconditionally by [Theorem 3.3](#). The rotational relative-equilibrium class is empty by the local Coriolis-flux and recentered concentration analysis in [Theorem 4.8](#). The degenerate stationary-hull class with retained local concentration is empty by the terminal local concentration theorem, through [Theorem 6.6](#). The generic terminal residual is empty, under the explicit realized critical-tail rigidity input, by the finite separated-family and indecomposability argument in [Theorem 10.2](#). Since the refined decomposition exhausts the residual class by definition, these four local exclusions imply that the residual class is empty. $\square$

Remark 11.1 (What is imported and what is proved here). The paper imports from Paper 0 only local positive concentration and the local Type I/Type II dichotomy, imports from Paper I only the Type I Seregin extraction and residual criterion, imports from Paper II only the results proved there, and imports from Paper III only the closed structured alternatives. The terminal active-locus extraction, diffuse-defect compactness, active path-space recurrence, and finite separated-family exclusion are not imported: they are implemented above by local concentration analysis. The realized critical-tail rigidity statement is the explicit local state-space input for the critical boundary compactification, not a hidden consequence of local CKN observables. The rotational relative-equilibrium class is not closed by a global Coriolis–Leray Liouville theorem for bounded profiles; it is closed by the localized flux identity and the same recentered local concentration analysis.

REFERENCES

- [1] G. Duran Ballester, *Local CKN concentration and the Type I/Type II dichotomy for Navier–Stokes*, preprint, 2026.
- [2] G. Duran Ballester, *Type I blow-up limits and residual closure for the three-dimensional Navier–Stokes equations*, preprint, 2026.
- [3] G. Duran Ballester, *Uniformly tight ancient Navier–Stokes dynamics and endpoint L^3 rigidity*, preprint, 2026.
- [4] G. Duran Ballester, *Structured ancient solution exclusions for Type I Navier–Stokes blow-up limits*, preprint, 2026.
- [5] D. Albritton and T. Barker, *On local Type I singularities of the Navier–Stokes equations and Liouville theorems*, J. Math. Fluid Mech. 21 (2019), Article 43.
- [6] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Comm. Pure Appl. Math. 35 (1982), 771–831.
- [7] L. Escauriaza, G. Seregin, and V. Sverak, *$L_{3,\infty}$ -solutions of Navier–Stokes equations and backward uniqueness*, Russian Math. Surveys 58 (2003), 211–250.
- [8] G. Koch, N. Nadirashvili, G. Seregin, and V. Sverak, *Liouville theorems for the Navier–Stokes equations and applications*, Acta Math. 203 (2009), 83–105.
- [9] Z. Lei, J. Ren, and X. Zhang, *A Liouville theorem for the axially symmetric Navier–Stokes equations*, J. Differential Equations 313 (2022), 213–238.
- [10] Z. Lei, Q. Zhang, and N. Zhao, *Improved Liouville theorems for axially symmetric Navier–Stokes equations*, J. Math. Fluid Mech. 19 (2017), 273–285.
- [11] J. Necas, M. Ruzicka, and V. Sverak, *On Leray’s self-similar solutions of the Navier–Stokes equations*, Acta Math. 176 (1996), 283–294.
- [12] G. Seregin, *A certain necessary condition of potential blow up for Navier–Stokes equations*, Comm. Math. Phys. 312 (2012), 833–845.